



Wwan Application Package

User Guide

V1.9

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Applicable OS & Model

No.	Applicable Model	Applicable OS	Description
1	RW350R	Chrome	--
2	RW135	Chrome	--
3	RW135R	Chrome	--

Change History

V1.9(2026-05-11)	Fix some description
V1.8(2026-03-18)	Fix some description
V1.7(2026-02-05)	Fix some description
V1.6 (2025-11-20)	Updated the format.
V1.5 (2025-09-16)	Fix some description
V1.4 (2025-08-10)	Fix some description
V1.3 (2025-07-15)	Fix some description
V1.2 (2025-06-30)	Add TASAR introduction Fix some description
V1.1 (2025-05-14)	Add Chrome installation steps
V1.0 (2025-03-21)	The Initial document

1 Introduction

1.1 Purpose of the Manual

Provide comprehensive usage guidelines for the Wwan Application Package, covering Engineer Tool, Telephony Tool, Noise Profiling Tool and Antenna Tuner Tool modules.

1.2 Target Users

Engineers, technical support staff, network operations teams, and roles requiring tool usage.

1.3 Manual Structure Overview

Organized by functional modules, with subsections for each tool and sub-features.

2 Tool Overview

2.1 Wwan Application Package Introduction

The Wwan Application Package contents following tools:

- Engineer Tool: Provides various debugging capabilities for WWAN modules.
- Telephony Tool: Provides logging and flashing capabilities for WWAN module sides to support analysis and troubleshooting.
- Noise Profiling Tool: Assists in the analysis of RF noise environments to support performance tuning and interference mitigation.
- Antenna Tuner Tool: Supports dynamic antenna tuning and matching to optimize RF performance across different frequency bands.

In addition to the tools provided, Wwan Application Package includes also Firmware Upgrade/Switch Services, Customization Services, PCIOT Services and Modem Authentication Services, which are selectively available based on the operating system.

* Please note that service-based features are not supported on Chrome OS.

The functional differences between Chrome OS in MTK platform and Qualcomm platform modules are shown in the following table:

Table 1. The function differences Chrome OS on RW350R and RW135/RW135R

Tool Name	RW350R	RW135/RW135R
Engineer Tool	RAT	RAT
	BodySAR	BodySAR
	OTA	OTA
	Thermal	--
	TAS AR	--
	Customizer	--
Telephony Tool	Capture Trace	Capture Trace
	Trace Setting	Trace Setting
	Modem Toolkit	Modem Toolkit
Noise Profiling Tool	Single Band Scan	Single Band Scan
	Multi Band Scan	Multi Band Scan
	Single Channel Scan	Single Channel Scan
Antenna Tuner Tool	Antenna Tuner	--

2.2 System Requirements

OS compatibility:

Linux Kernel version requires 6.13 and above (support for PCIe device fastboot switch function starting from 6.12, support for NPT port starting from 6.13)

2.3 Installation & Uninstallation

Chrome OS

- Installation Steps:

1. Open Shell Window in Chrome and Enter the dev console
2. Press Ctrl + Alt + T to open dev console in new tab.
3. On the command line, type following commands in sequence:
 - shell
 - sudo su



If this command cannot enter the root user, you can try the following two methods

- sudo -i
- ssh: ssh root@127.0.0.1

If the above method cannot enter the root user, please follow the steps below

- ctrl+alt+refresh key to enter the command line interface
- root
- /usr/libexec/debugd/helpers/dev_features_rootfs_verification
- reboot, after booting, repeat steps 1 and 2 to enter the root permission interface
- /usr/libexec/debugd/helpers/dev_features_ssh
- passwd, set the password to test0000
- reboot

4. Password: test0000 (the password is implicit, please enter the password, if the console show the local host path, it means the password is correct)
5. Copy installer with below command:
 cp /path/to/wwan-service-tool-generic-x.x.x.run /usr/local/
 Replace /path/to/wwan-service-tool-generic-x.x.x.run with the actual path to your installer file.
6. Grant execution permissions to the file if needed:
 chmod +x /usr/local/wwan-service-tool-generic-x.x.x.run
7. Execute the installer to begin the setup:
 /usr/local/wwan-service-tool-generic-x.x.x.run

- Uninstallation Steps:

1. Open Shell Window in Chrome and Enter the dev console
2. Enter the root user using method above
3. Execute the script to uninstall:

```
/usr/local/opt/waf/script/uninstall.sh
```

* In some cases, the WWAN application package may not be fully uninstalled. Please follow the steps below to manually remove any remaining components.

- cd /etc/init
- stop wwan_tool
- stop wwan_func
- stop wwan_dev
- rm -rf "wwan_dev.conf"
- rm -rf "wwan_func.conf"
- rm -rf "wwan_tool.conf"
- rm -rf /usr/local/opt/waf



If the following message shown in Figure 2-1 appears after installing wwan_tool, it indicates that system memory is insufficient, the installation has failed, and web pages cannot be opened. You need to uninstall the tool to manually clean up the environment, and then reinstall it.

```
chromeos_metadata_partition empty stateful_partition
uldrnite-rev3 /mnt # cd stateful_partition/
uldrnite-rev3 /mnt/stateful_partition # df -h .
Filesystem      Size  Used Avail Use% Mounted on
/dev/mapper/defaultkey_encrypted 49G  28G  27G  43% /mnt/stateful_partition
uldrnite-rev3 /mnt/stateful_partition # cd /usr/local
uldrnite-rev3 /usr/local # ./opt/waf/script/uninstall.sh
wwan_tool stop/waiting
wwan_func stop/waiting
wwan_dev stop/waiting
Deleting the waf service file: /etc/init/wwan_dev.conf /etc/init/wwan_dev.conf
Deleting the waf directory: /usr/local/opt/waf
uninstall wwan service success!
uldrnite-rev3 /usr/local # ./wwan-service-tool-generic-1.0.14.run
[*] Extracting installer to /usr/local/tmp/wwan_tmp.8392 ...
[*] Installing WWAN Service Tool version 1.0.14 for x86_64 ...
Directory /usr/local/opt/waf/script does not exist. No action taken.
Extracting files to /usr/local/opt...
install successfully, install path /usr/local/opt
Running installation script...
Directory /usr/local/opt/waf/etc/init exists. Copying files to /etc/init...
mv: error writing '/etc/init/wwan_tool.conf': No space left on device
wwan_dev start/running, process 8422
wwan_func start/running, process 8425
wwan_tool start/running
wwan_dev, wwan_func, and wwan_tool started successfully!
install wwan service success!
uldrnite-rev3 /usr/local # ./opt/waf/script/uninstall.sh
wwan_tool stop/waiting
wwan_func stop/waiting
wwan_dev stop/waiting
Deleting the waf service file: /etc/init/wwan_dev.conf /etc/init/wwan_dev.conf
Deleting the waf directory: /usr/local/opt/waf
uninstall wwan service success!
uldrnite-rev3 /usr/local # df -h
Filesystem      Size  Used Avail Use% Mounted on
```

Figure 2-1. Installation error message

3 Launch Toolkit

3.1 Launch Toolkits

3.1.1 Launch Toolkit main page

Wwan Application Package Toolkit is built on a browser-WebSocket server architecture, with the user interface rendered directly in a web browser.

It supports modern browsers including Chrome, Edge, Firefox, Opera, and Brave.

- **Windows and Linux**

On Windows and Linux platforms, the Toolkit launch icon is available in the system's start menu. Users can double-click the icon to launch the Toolkit, which will automatically open the default web browser and load the Toolkit main page.

If the browser is already running, the Toolkit will open a new window within the existing browser instance to display the main page.

- **Chrome OS**

On Chrome OS, the Toolkit runs as a background service without a desktop launch icon. Users must manually open the Toolkit by navigating to <http://localhost:8000> or <http://127.0.0.1:8000> in their web browser.

3.1.2 Launch individual tool

Wwan Application Package Toolkit is an integrated toolchain platform that provides a unified interface for accessing various tools.

The main page displays all available tools in a single view. Currently supported tools include: Engineer Tool, Telephony Tool, Noise Profiling Tool, and Antenna Tuner Tool. Among them, The Chrome OS module on the Qualcomm platform does not support the Antenna Tuner Tool.

Users can access each tool by clicking on its corresponding icon, which opens the tool in a new browser tab or window.

Multiple tools can run simultaneously, provided there are no functional conflicts. For example, while a flashing operation is in progress, modem-related functions will be temporarily restricted to ensure system stability and avoid interference.

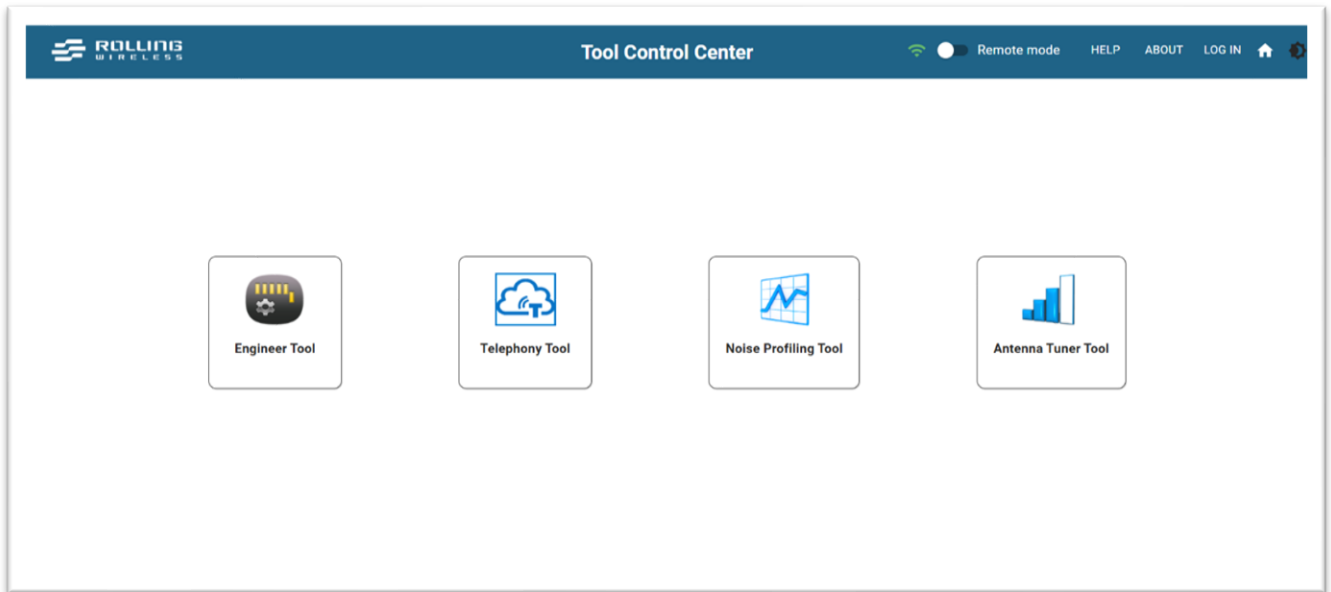


Figure 3-1. Toolkit Main Page of MTK platform

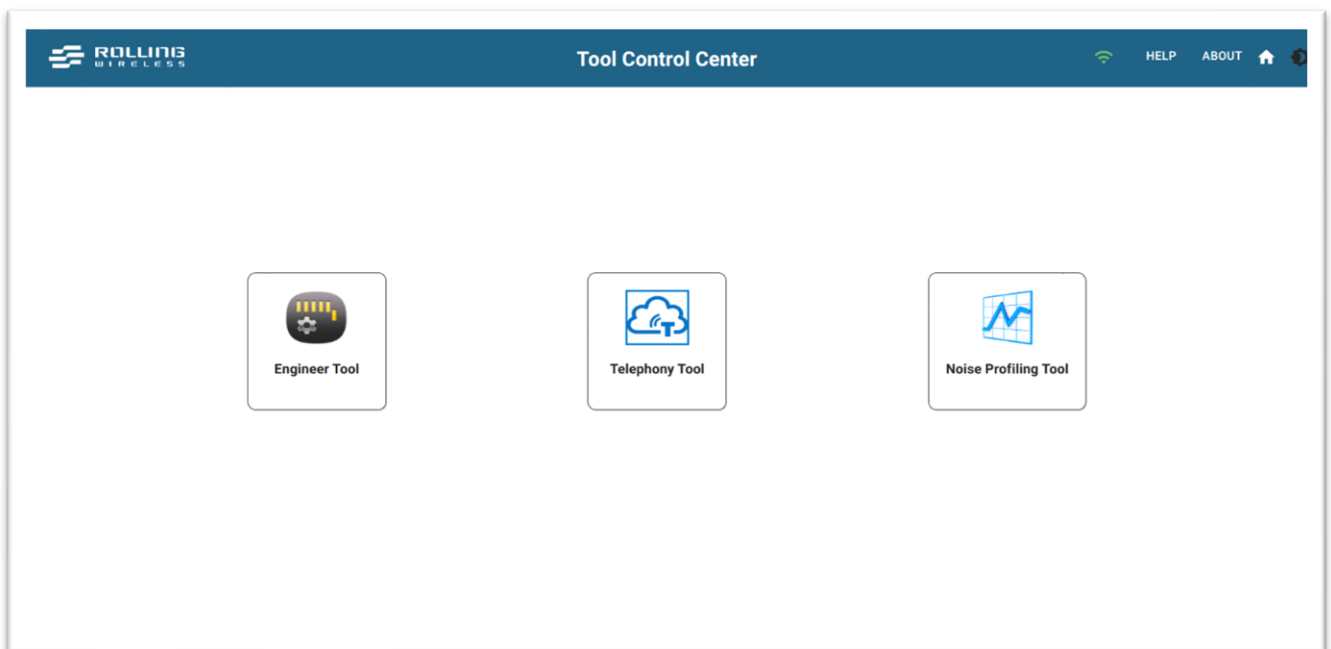


Figure 3-2. Toolkit Main Page of Qualcomm platform

3.2 Common Functions

At the top of each tool interface, three common functions are displayed in the header: Connection, Device Information, and AT Sender.

Each section can be expanded or collapsed by clicking on its panel header, allowing users to customize the view based on their workflow and preferences.

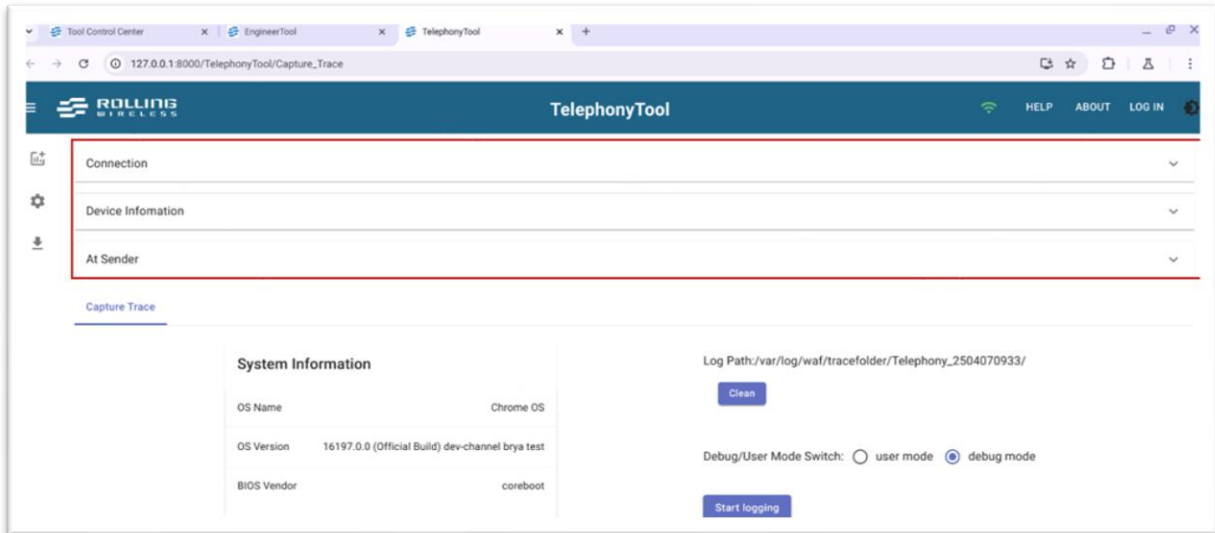


Figure 3-3. Collapsed Common Functions

1. Connection

Wwan Application Package Toolkit provides the flexibility to switch connection ports based on user requirements.

All available ports are listed in a drop-down combo box, allowing users to easily select the desired port. Once selected, AT commands are automatically routed through the chosen port, ensuring seamless communication with the connected device.



When both the MBIM port and the GNSS Test port are available, selecting the GNSS port will not be retained after a webpage refresh. The Connection will automatically revert to the MBIM port.

2. Device Information

Basic tool-related information is available in the Device Information section.

The specific details displayed may vary depending on the underlying modem platform, ensuring that relevant and platform-specific data is presented to the user.

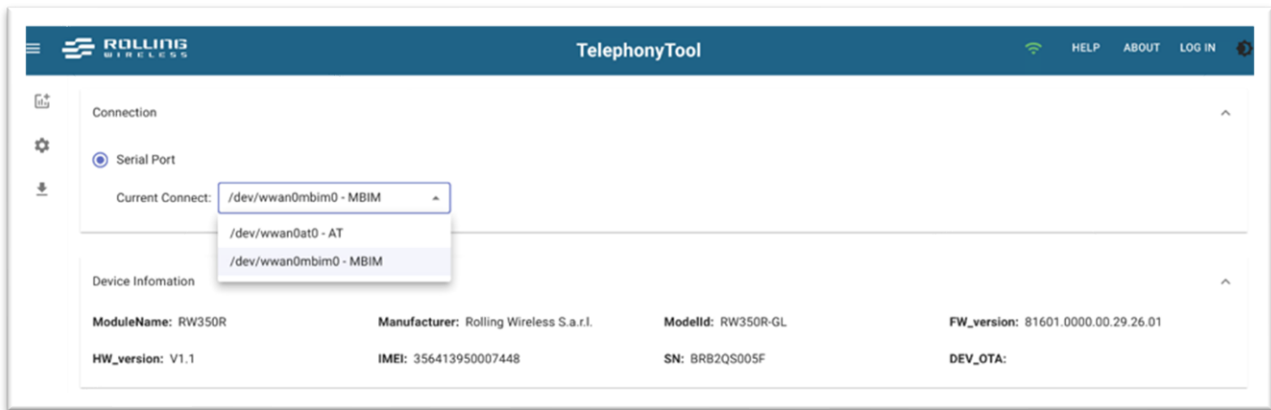


Figure 3-4. Expanded Connection and Device Information Section

If the toolkit detects an abnormal modem status, all device information fields will display as "Unknown," and an alert message will appear.

If the issue is due to a temporary anomaly, users can attempt to restore the device information by clicking the "Retry" hyperlink provided in the alert.

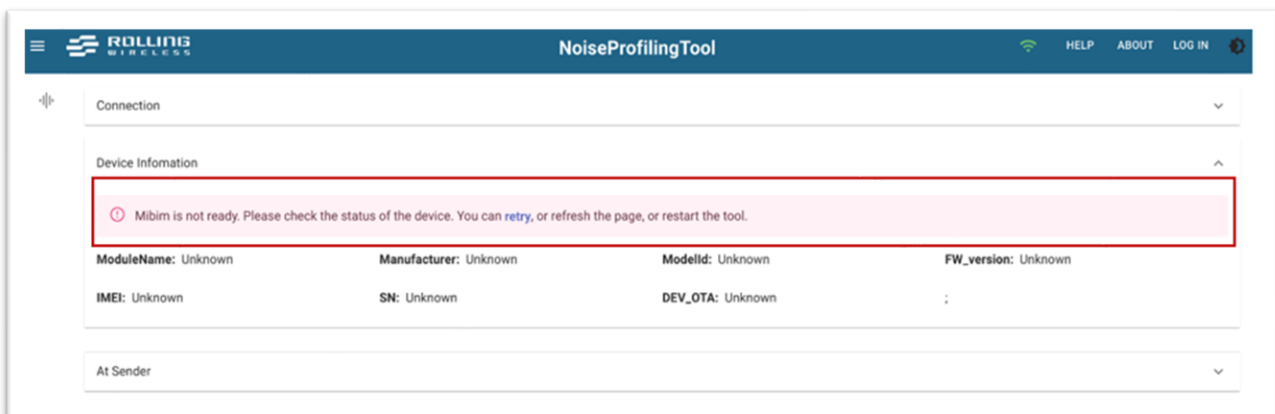


Figure 3-5. Device Abnormal Status

3. AT sender

Wwan Application Package Toolkit includes an AT command interface with a collection of frequently used AT commands readily available.

Users can either select a predefined command from the list or manually enter a custom command in the input field and press the Enter key to send it.

Figure 3-6. Expanded AT Sender Section

Custom commands are automatically saved in the list, allowing for easy reuse in future sessions.

Figure 3-7. Custom commands saved in list

4 Engineer Tool User Guide

4.1 Module Overview

Engineer Tool is a tool used during the development and debugging phase, mainly divided into the following functions:

- RAT
- Body Sar
- Thermal (MTK platform only)
- OTA
- TA SAR (MTK platform only)
- Customizer (MTK platform only)

* The available UI functions vary depending on the platform.

4.2 RAT

RAT unit is used for field testing.

4.2.1 Use Prerequisites

The outfield function test requires inserting a valid SIM card and connecting the antenna.

4.2.2 Use Instructions

The following figure shows the UI of the Engineer Tool RAT.

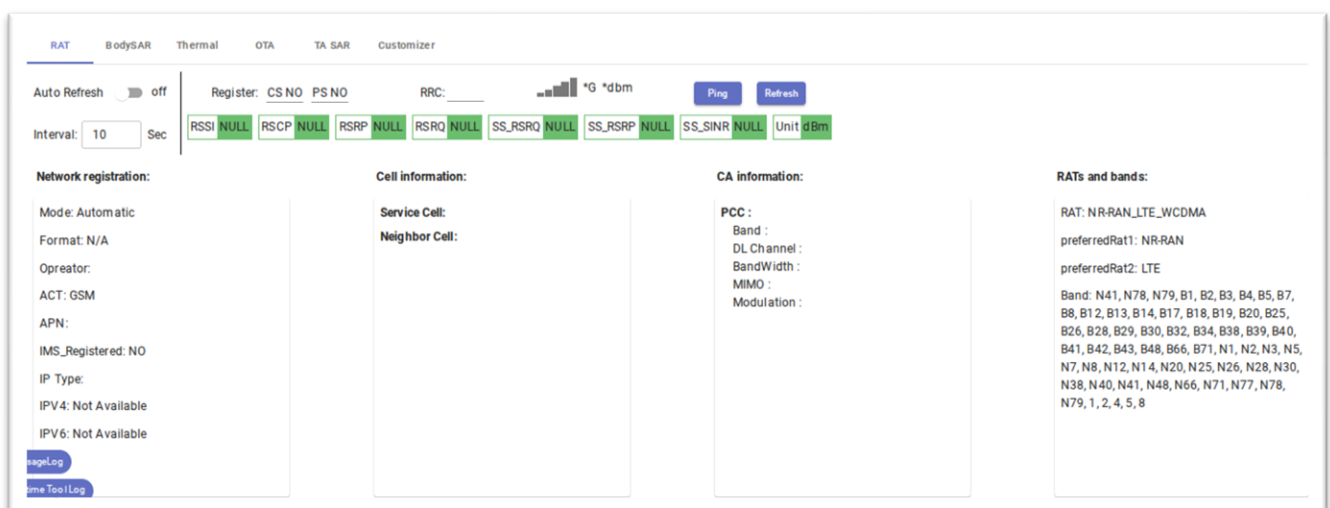


Figure 4-1. Engineer Tool RAT

4.2.2.1 Auto Refresh



Figure 4-2. Auto Refresh

1. RAT provides the ability to automatically refresh interface information, which is turned off by default.
 2. When you click the Auto Refresh slide button in the figure above, the feature is enabled.
 3. RAT also provides the ability to customize automatic refresh intervals, which are 10 seconds by default.
- When auto-refresh is enabled, the auto-refresh interval settings are grayed out and cannot be set.

4.2.2.2 Refresh Manually

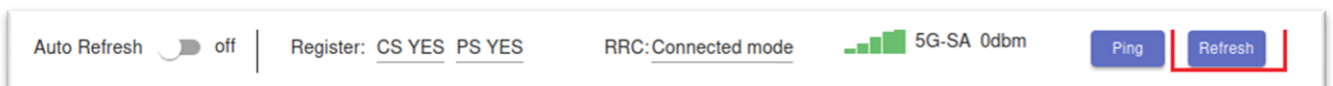


Figure 4-3. Refresh button

Clicking the Refresh button refreshes everything on the UI.

4.2.2.3 Signal Strength



Figure 4-4. Signal Strength

Displays the current rssi, rscp, rsrq, rsrp, ss_rsrq, ssrsrp, ss_sinr, communication type (2G/3G/4G/5G) and signal strength in dBm.

4.2.2.4 Register Status

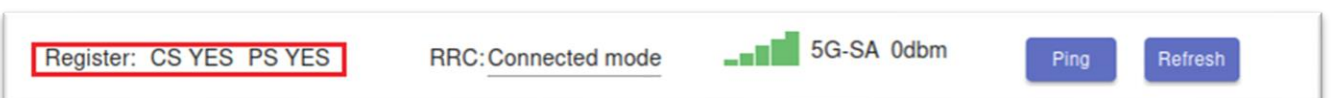


Figure 4-5. Register Status

Register status contains three statuses: **Registered**, **Not Registered**, and **Unknown**. Yes means

registered and No means not Registered.

4.2.2.5 RRC Status

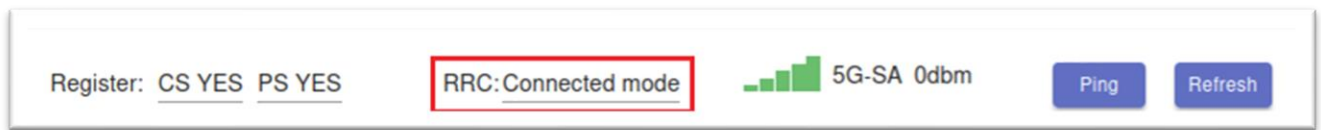


Figure 4-6. RRC Status

RRC status contains three statuses: **Idle mode**, **Connected mode**, and **Unknown**.

4.2.2.6 Network Registration



Figure 4-7. Network Registration

Network registration information display:

- Mode: manual, automatic, deregister from network, manual/automatic
- Format: long format alphanumeric, short format alphanumeric, numeric
- Operator: The network operator identification number. For example, 46000 represents China Mobile.
- ACT: GSM, GSM Compact, UTRAN, GSM w/EGPRS, UTRAN w/HSDPA, UTRAN w/HSUPA, UTRAN w/HSDPA and HSUPA, E-UTRAN, EC-GSM-IoT (A/Gb mode), E-UTRAN (NB-S1 mode), E-UTRA connected to a 5GCN, NR connected to a 5GCN, NG-RAN, E-UTRA-NR dual connectivity
- PDP Type: IP, IPV6, IPV4V6, PPP
- IMS: IP Multimedia Subsystem. "Yes" indicates that it is registered, and "No" indicates that it is not registered.
- APN: Access Point Name. For example, "cmnet".
- IP: The IP address consists of IPV4, IPV6, IPV4 is represented by a decimal string, and IPV6 is

represented by a hexadecimal string.

4.2.2.7 Cell Information

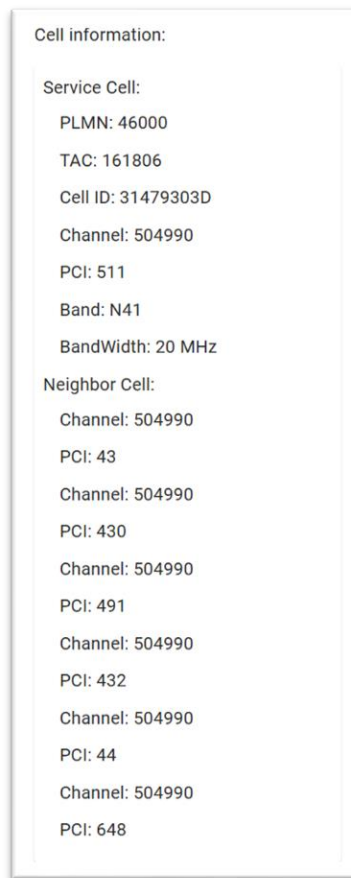


Figure 4-8. Cell Information

Displays cell information for service cell and neighbor cell:

- PLMN: It consists of MCC (Mobile Country Code) + MNC (Mobile Network Code).
- TAC: Tracking Area Code
- Cell ID: The range is 0-0xFFFFFFFF
- Channel: Absolute Radio Frequency Channel Number
- PCI: Physical Cell ID
- Band: Band value
- Bandwidth: The range is 0-255

4.2.2.8 CA Information

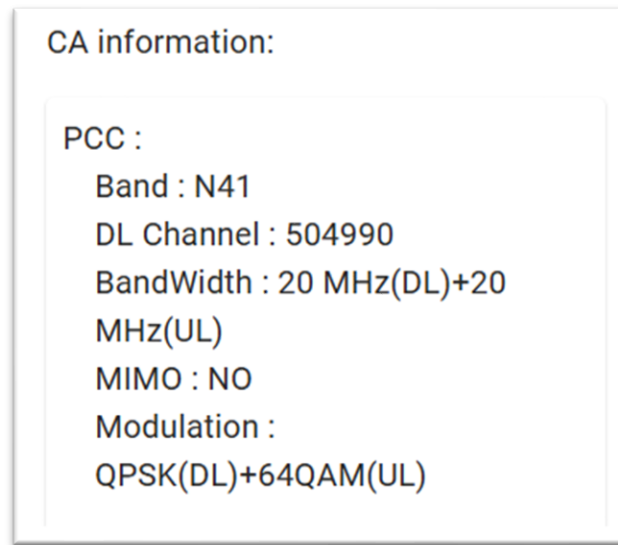


Figure 4-9. CA Information

Displays carrier Aggregation information, including PCC and SCC data content:

- Band: Band value
- DL Channel: Radio Frequency Channel Number
- Bandwidth: The range is 0–255
- MIMO: Multi Input Multi Output
- Modulation: BPSK, QPSK, 16QAM, 64QAM, 256QAM, 1024QAM, UNKNOWN
- CA Status: State of the secondary cell. There are two states: configured and deactivated, configured and activated.

4.2.2.9 RATs and Bands

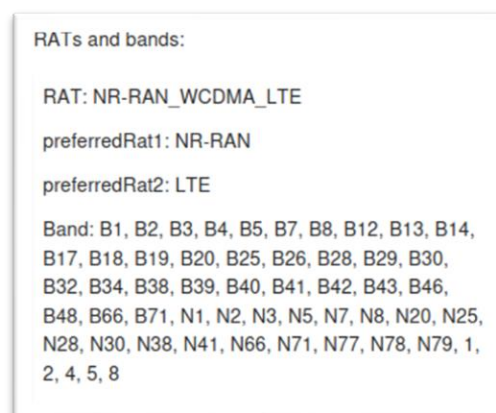


Figure 4-10. RATs and Bands

Displays RATs and bands information, including RAT, preferred ACT1, preferred ACT2, and band values:

- RAT: Radio Access Technology
- PreferredAct1: First access option
- PreferredAct2: Second access option
- Bands: Configured bands

4.3 Body SAR

The SAR Tuning function is used by Rolling customers for body SAR debugging. In this unit, the user can easily configure the Body SAR table of the module in this unit. OTA—>MAX Tx Power needs to be reset when testing BodySAR.

Note: Prior to executing these testing functions on Windows or Linux systems, please ensure that the **Service Flash** option is disabled from the **Telephony Tool Download** page.

4.3.1 The Main Interface

4.3.1.1 Supported Function

This unit supported functions are as follows:

- Body SAR switch
- Body SAR table configuration
- Command status

4.3.1.2 Main Interface

The main interface is as follows:

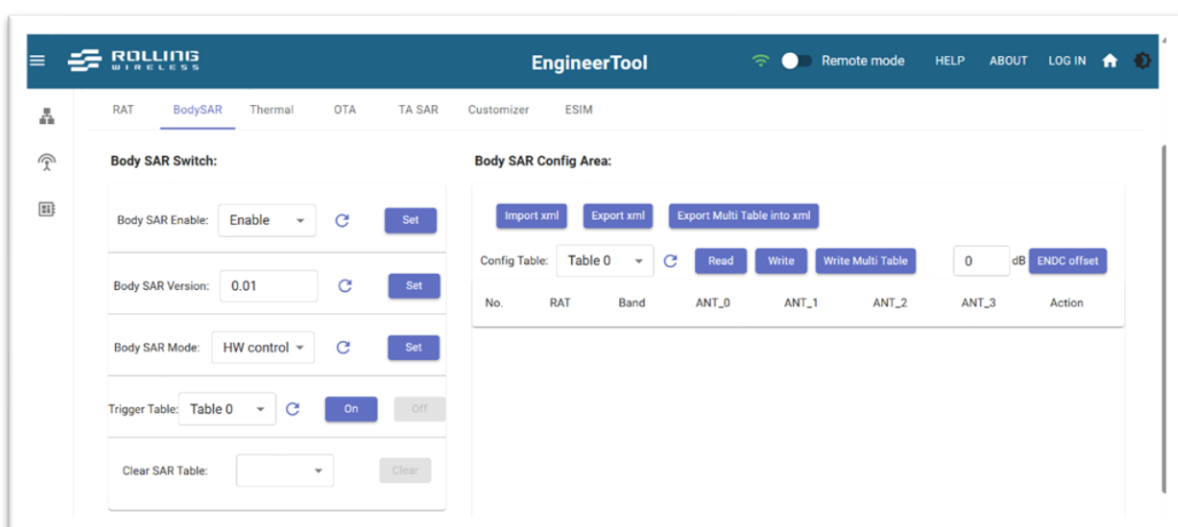


Figure 4-11. Main Interface

4.3.2 Body SAR Switch

4.3.2.1 Body SAR Enable

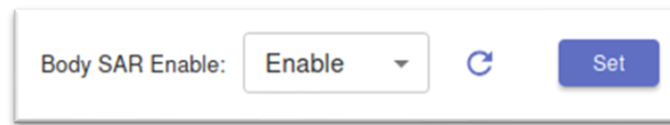


Figure 4-12. Body Sar Enable

This area is the enable switch of the Body SAR function. When the setting is enabled, the entire UI can be operated; when the setting is disabled, all items are inoperable except for the Body SAR Enable.

The **Set Body SAR Enable** drop-down box is used to display the current Body SAR Enable status. Click the refresh button on the right to refresh the status.

Click the **Set** button to set Body SAR Enable to **Enable** or **Disable**. After clicking, the current enable status of the module will be automatically read and displayed. If it cannot be modified, the setting is unsuccessful.

4.3.2.2 Body SAR Mode

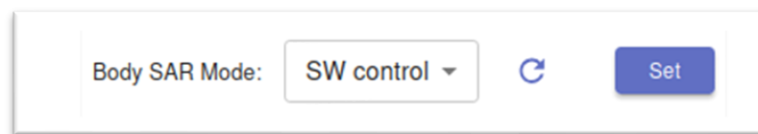


Figure 4-13. Body SAR Mode

This area is to set the control mode of Body SAR, including hardware control (HW control) and software control (SW control)

Set Body SAR Mode drop-down box is used to display the current control mode. You can click the refresh button on the right to read the value from the module.

Click the **Set** button to set the Body SAR Mode to **HW control** or **SW control**. After clicking, the current mode of the module will be automatically read and displayed. If it cannot be modified, the setting is unsuccessful.

4.3.2.3 Trigger Table

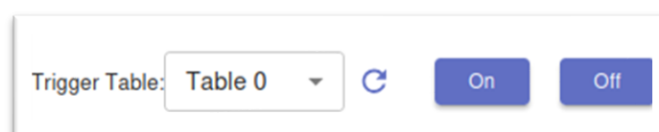


Figure 4-14. SAR Trigger Table

This area is to set the currently used Table and enable or disable the trigger table.

1. The **Set Trigger Table** drop-down box will display the currently used Table. Click the refresh button on the right to read the value from the module.
2. Click the **Set Trigger Table** drop-down box to set the Body SAR Table, the current Table of the module will be automatically read and displayed. If it cannot be modified, the setting is unsuccessful.
3. Click **ON** Button to enable the trigger table.
4. Click **OFF** Button to disable the trigger table.

4.3.2.4 Clear Table

This area is used to clear the SAR value of the selected table. If the cleared table index is enabled, the antenna will work at the maximum power.



Figure 4-15. Clear SAR Table

4.3.3 Body SAR Quick Set

Body SAR Quick Set sets the maximum transmit power corresponding to all mobile communication technologies (3G/4G/5G) and frequency bands supported by the module one by one by reading the xml file; if the setting item is not supported by the module, the setting fails. The content of a single file includes tech (mobile communication technology), band (frequency band) and power (maximum transmit power).

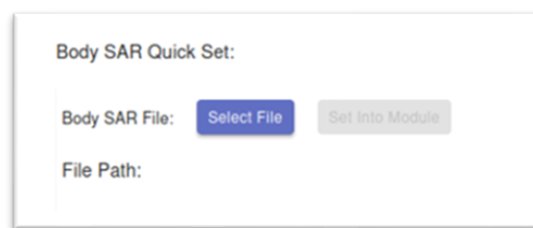


Figure 4-16. Body SAR Quick Set

1. Click **Select File** to open the file selection dialog box. The required XML document can be selected from dialog box.
2. After selecting the file, it will check whether the content of the file is valid. If it is valid, the full path of the file will be displayed in the text, as shown below.
3. Click the **Set Into Module** button to set the content of the file to the module line by line. During the setting process, the entire UI is grayed out and cannot be operated. The UI will not return to normal until all data is set or the setting fails.

4.3.4 Body SAR Table Configuration

4.3.4.1 Export Multiple Tables

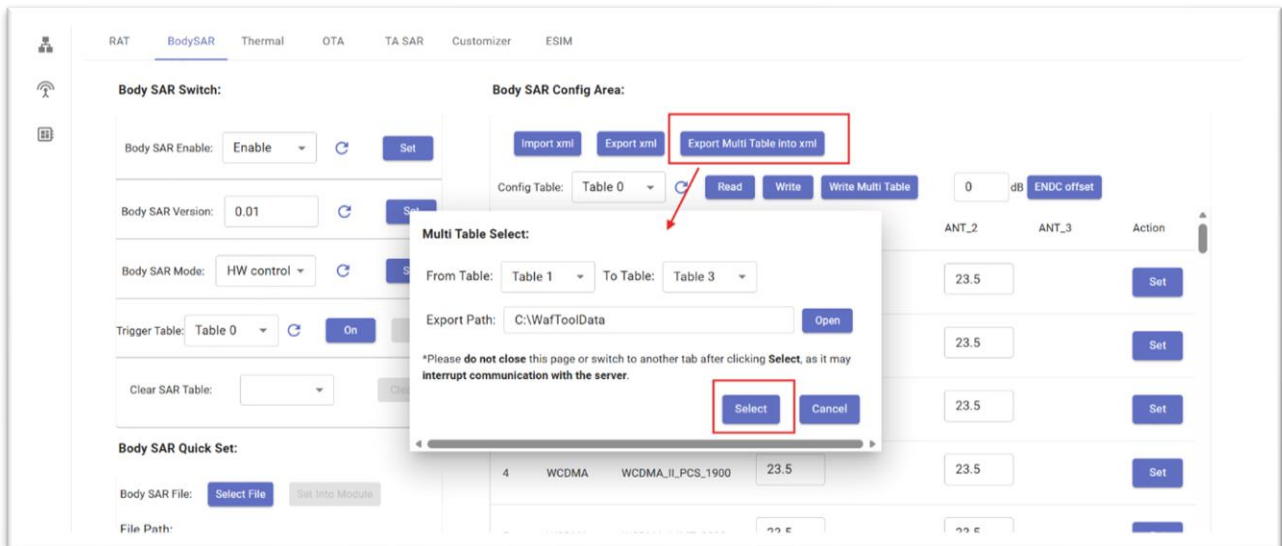
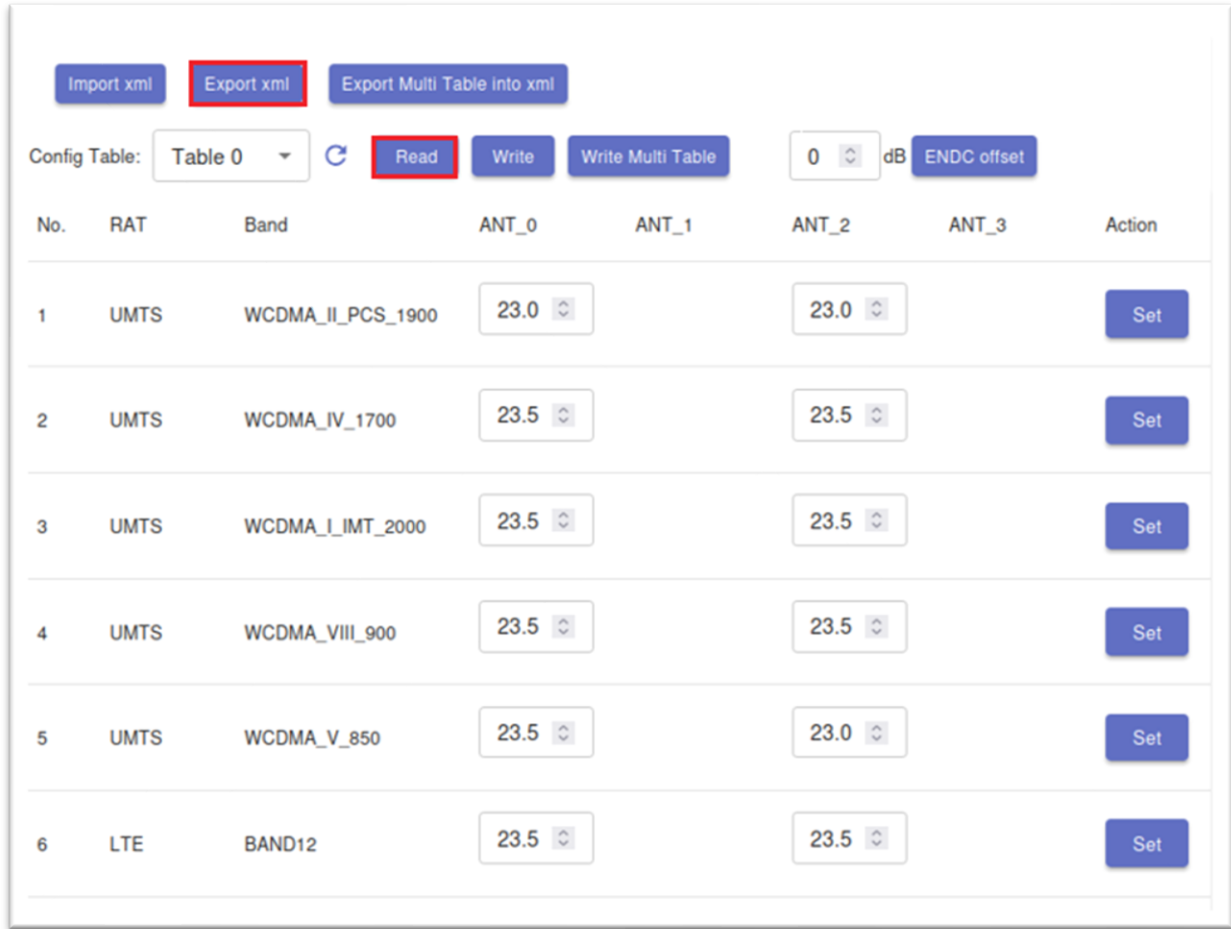


Figure 4-17. Export multiple Tables

For example, click **Export Multi Table into xml**, Select the table to be exported and click the **Select** button.

4.3.4.2 Read & Write Body SAR



No.	RAT	Band	ANT_0	ANT_1	ANT_2	ANT_3	Action
1	UMTS	WCDMA_II_PCS_1900	23.0		23.0		Set
2	UMTS	WCDMA_IV_1700	23.5		23.5		Set
3	UMTS	WCDMA_I_IMT_2000	23.5		23.5		Set
4	UMTS	WCDMA_VIII_900	23.5		23.5		Set
5	UMTS	WCDMA_V_850	23.5		23.0		Set
6	LTE	BAND12	23.5		23.5		Set

Figure 4-18. Read and Set Body Sar Table

This item can import table, export table, edit table, read and set table.

UI introduction:

1. Column 1 of the table indicates the index of the Body SAR setting item;
2. Column 2 of the table indicates the type of mobile communication technology, including WCDMA, LTE, NR and others.
3. Column 3 of the table indicates the frequency band value
4. Column 4 to Column 7 of the table indicate the power values of different antennas (unit: dBm);

Function introduction:

1. Click the Set behind each row to set the data in the corresponding row of the module as row-change data, which is a single row modification.
2. Click the Read from Module to read the Body SAR data of the current table from the module.
3. Click the Set All into Module to synchronously set the current table data row by row to the corresponding row of the module. If there is a row in the middle of the setting fails, all subsequent settings will be aborted. After clicking the Set into Module button, the entire interface will be

inoperable until the setting is completed or the setting is abnormally terminated, and the entire UI returns to normal.

4. Click Import to import an xml file to the table.
5. Click the Export to export the table data to an xml file; the file path is fixed to the Body SAR folder under the application path, and the export file naming rule is Body SARFile_time.xml.
6. Click the Export as Excel to export the table data to an excel file; the file path is fixed to the Body SAR folder under the application path, and the export file naming rule is Body SARFile_Index.xml.

4.3.4.3 Write Multi Body SAR Table

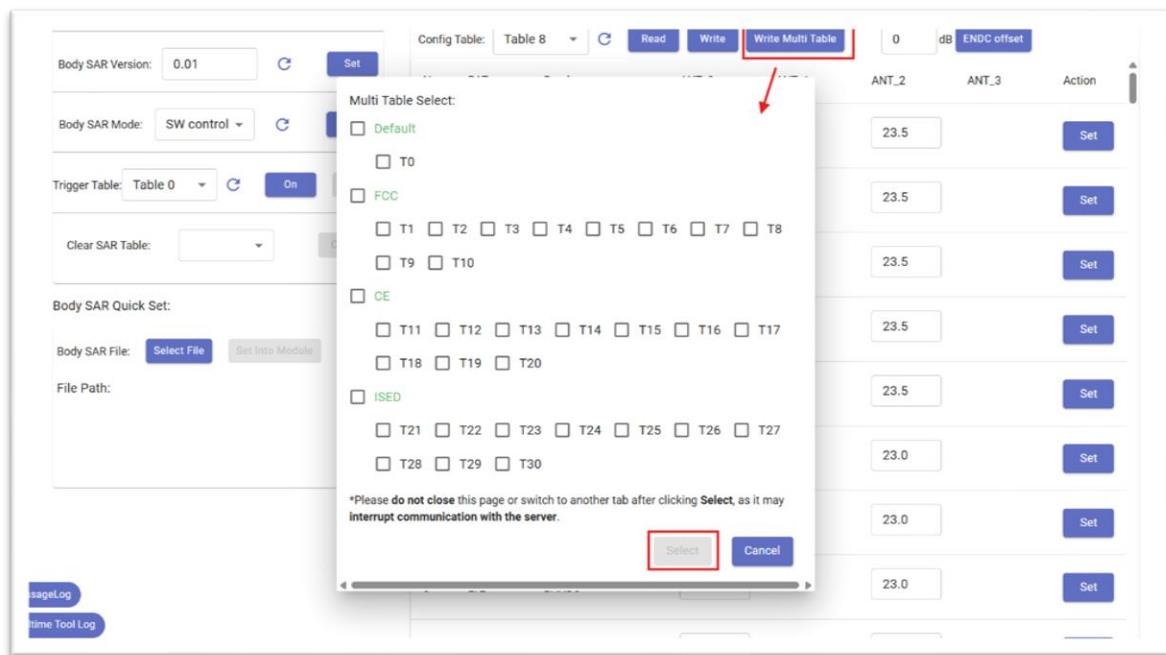


Figure 4-19. Set Multi Body Sar Table

1. Click the Write Multi Table button to show table maps.
2. Select the Body SAR table you want to write.
3. Click the Select button to write Body SAR value.

4.3.4.4 ENDC offset

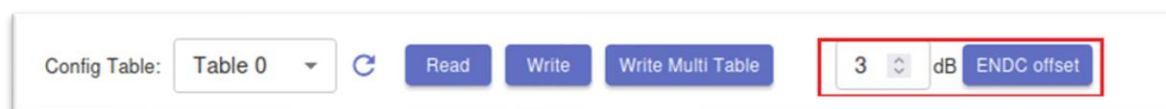
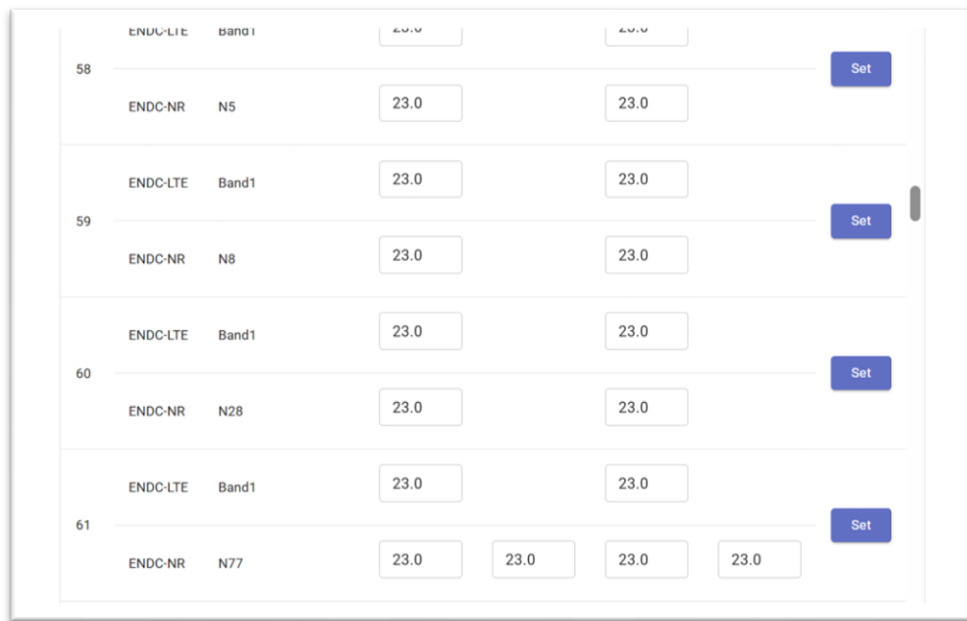


Figure 4-20. ENDC Offset setting

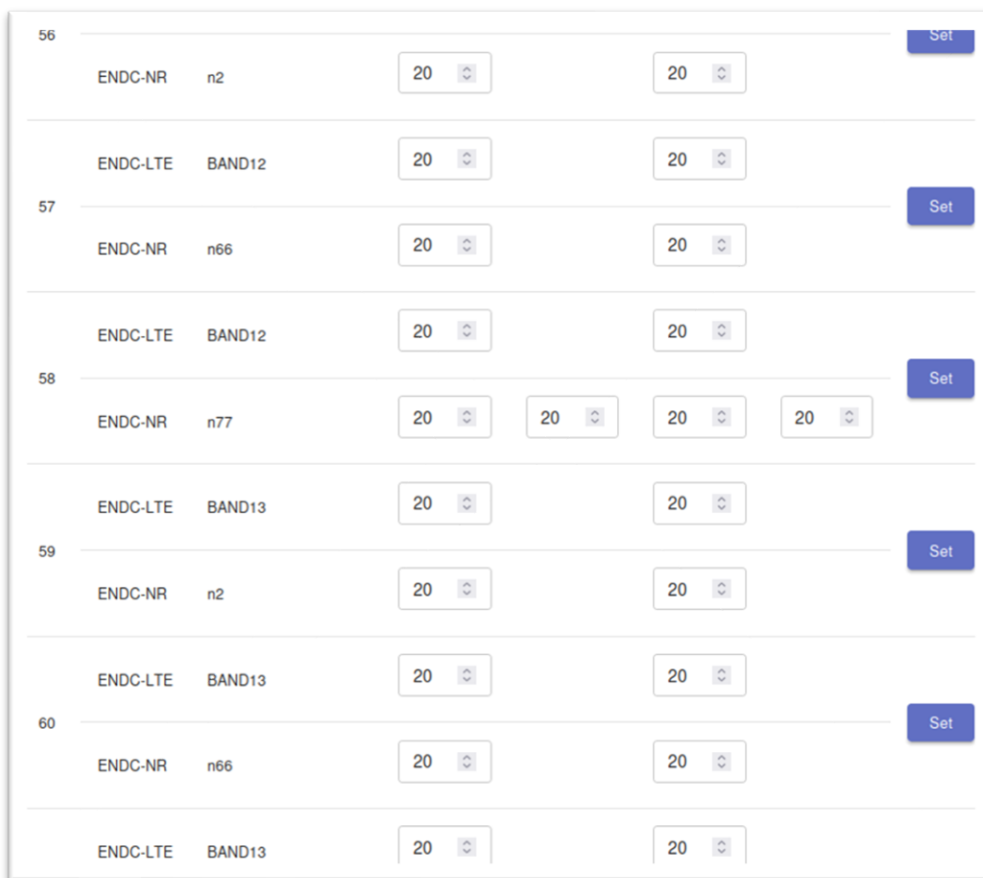
Click the **ENDC offset** button to change the value of the ENDC in batches (Default is 0 dB). The effect of the modification is shown in the following figure.

Whether 3 or -3 is fill in, it is reduced by 3dB based on the current single band value.



Band	ENDC-LTE	Band	ENDC-NR	Offset (dB)	Set
58	ENDC-LTE	Band1	N5	23.0	Set
59	ENDC-LTE	Band1	N8	23.0	Set
60	ENDC-LTE	Band1	N28	23.0	Set
61	ENDC-LTE	Band1	N77	23.0	Set

Figure 4-21. Before ENDC Offset setting



Band	ENDC-LTE	Band	ENDC-NR	Offset (dB)	Set
56	ENDC-LTE	BAND12	n2	20	Set
57	ENDC-LTE	BAND12	n66	20	Set
58	ENDC-LTE	BAND13	n77	20	Set
59	ENDC-LTE	BAND13	n2	20	Set
60	ENDC-LTE	BAND13	n66	20	Set

Figure 4-22. After ENDC Offset setting

4.3.5 Configuration Suggestion

4.3.5.1 Table Index Allocation

4.3.5.2 SW Mode

In SW mode, table configuration methods can refer to the following table.

SW SAR Table Format								
HW SKU	Regulatory Mode	Device Mode		Antenna Sensors			SAR Table	
		Mode	BIOS	Sensor 1	Sensor 2	ANT TX 1	ANT TX 2	Host Send to MD
NA SKU	FCC	LapTop	Mode = 1	0	0	Full	Full	0
			Mode = 1	0	1	Full	SAR ANT2 CONFIG1 1	1
			Mode = 1	1	0	SAR ANT1 CONFIG1 1	Full	2
		Tablet	Mode = 1	1	1	SAR ANT1 CONFIG1 1	SAR ANT2 CONFIG1 1	3
			Mode = 0	0	0	Full	Full	4
			Mode = 0	0	1	Full	SAR ANT2 CONFIG1 2	5
	CE	LapTop	Mode = 0	1	0	SAR ANT1 CONFIG1 2	Full	6
			Mode = 0	1	1	SAR ANT1 CONFIG1 2	SAR ANT2 CONFIG1 2	7
			Mode = 1	0	0	Full	Full	8
		Tablet	Mode = 1	0	1	Full	SAR ANT2 CONFIG2 1	9
			Mode = 1	1	0	SAR ANT1 CONFIG2 1	Full	10
			Mode = 1	1	1	SAR ANT1 CONFIG2 1	SAR ANT2 CONFIG2 1	11
			Mode = 0	0	0	Full	Full	12
			Mode = 0	0	1	Full	SAR ANT2 CONFIG2 2	13
			Mode = 0	1	0	SAR ANT1 CONFIG2 2	Full	14
EU SKU	FCC	LapTop	Mode = 0	1	1	SAR ANT1 CONFIG2 2	SAR ANT2 CONFIG2 2	15
			Mode = 1	0	0	Full	Full	16
			Mode = 1	0	1	Full	SAR ANT2 CONFIG1 1	17
		Tablet	Mode = 1	1	0	SAR ANT1 CONFIG1 1	Full	18
			Mode = 1	1	1	SAR ANT1 CONFIG1 1	SAR ANT2 CONFIG1 1	19
			Mode = 0	0	0	Full	Full	20
	CE	LapTop	Mode = 0	0	1	Full	SAR ANT2 CONFIG1 2	21
			Mode = 0	1	0	SAR ANT1 CONFIG1 2	Full	22
			Mode = 0	1	1	SAR ANT1 CONFIG1 2	SAR ANT2 CONFIG1 2	23
		Tablet	Mode = 1	0	0	Full	Full	24
			Mode = 1	0	1	Full	SAR ANT2 CONFIG2 1	25
			Mode = 1	1	0	SAR ANT1 CONFIG2 1	Full	26
			Mode = 1	1	1	SAR ANT1 CONFIG2 1	SAR ANT2 CONFIG2 1	27
			Mode = 0	0	0	Full	Full	28
			Mode = 0	0	1	Full	SAR ANT2 CONFIG2 2	29
			Mode = 0	1	0	SAR ANT1 CONFIG2 2	Full	30
			Mode = 0	1	1	SAR ANT1 CONFIG2 2	SAR ANT2 CONFIG2 2	31

Figure 4-23. Sar Table Configuration

4.3.5.3 HW Mode

In HW mode, table configuration methods can refer to the following table.

Note: Do not support in current version.

HW SAR Mapping Table Rule								
Regulatory Mode	Device Mode		DPR HW PIN number			SAR Table		SAR Send to MD
	Mode	Mode PIN	DPR1	DPR2	ANT TX 1	ANT TX 2	Scenario index	
FCC	Mode A	1	0	0	Full	Full	0	
		1	0	1	Full	SAR ANT2 CONFIG1 1	1	
		1	1	0	SAR ANT1 CONFIG1 1	Full	2	
		1	1	1	SAR ANT1 CONFIG1 1	SAR ANT2 CONFIG1 1	3	
	Mode B	0	0	0	Full	Full	4	
		0	0	1	Full	SAR ANT2 CONFIG1 2	5	
		0	1	0	SAR ANT1 CONFIG1 2	Full	6	
		0	1	1	SAR ANT1 CONFIG1 2	SAR ANT2 CONFIG1 2	7	
CE	Mode A	1	0	0	Full	Full	8	
		1	0	1	Full	SAR ANT2 CONFIG2 1	9	
		1	1	0	SAR ANT1 CONFIG2 1	Full	10	
		1	1	1	SAR ANT1 CONFIG2 1	SAR ANT2 CONFIG2 1	11	
	Mode B	0	0	0	Full	Full	12	
		0	0	1	Full	SAR ANT2 CONFIG2 2	13	
		0	1	0	SAR ANT1 CONFIG2 2	Full	14	
		0	1	1	SAR ANT1 CONFIG2 2	SAR ANT2 CONFIG2 2	15	
		0	1	1	SAR ANT1 CONFIG2 2	SAR ANT2 CONFIG2 2	15	
		0	1	1	SAR ANT1 CONFIG2 2	SAR ANT2 CONFIG2 2	15	

Figure 4-24. Sar Table Configuration Methods



PIN definition depends on OEM requirements and design.

4.3.6 SAR mechanism in UL MIMO/ULCA mode

In UL MIMO mode:

the power control used EPS mechanism, two Tx stream will have the similar Tx power and control the total max Tx power as the single band Max Tx power w/o UL MIMO. (e.g. n41 HPUE max power is 26dBm w/o UL MIMO, n41 UL MIMO with HPUE total power will also be 25dBm<for UL MIMO only support CP-OFDMA, the max power will back off 1dB follow the 3Gpp, so the real max power is 25dBm>, each Tx stream actual power will be ~22dBm), so the UL MIMO will call the single band w/o UL MIMO SAR setting directly, no dedicated SAR configure for UL MIMO.

In UL CA mode:

For intra-band non-contiguous ULCA, module not support it, so no need to configure SAR,

For intra-band contiguous ULCA, two CC work in the same ANT and the total max Tx power is same as the single band, FW will call the single band SAR setting directly, no dedicated SAR configure

For inter band UL CA, no dedicated SAR configure, the FW have one setting to enable extra 3dB back off, so the situation is:

1. Disable extra 3dB back off, each Tx stream will call each single band SAR configure.
e.g. assume single band mode, B1 SAR setting in ANT2 is 18dBm, B8 SAR setting in ANT0 is 19dBm.
The ULCA 1A(ANT2)-8A(ANT0) SAR will be: B1 use 18dBm, B8 use 19dBm
2. Enable extra 3dB back off, each Tx stream will call each single band SAR configure value -3dB.
e.g. assume single band mode, B1 SAR setting in ANT2 is 18dBm, B8 SAR setting in ANT0 is 19 dBm.
The ULCA 1A(ANT2)-8A(ANT0) SAR will be: B1 use 15 dBm(18-3), B8 use 16dBm(19-3)

4.3.7 SAR Up threshold setting for TDD band

Device need use the max UL Tx slot configure to do the tuning, then set the Tx max power limitation in SAR mode:

4.3.7.1 For NR

The 3Gpp defined that max UL Tx slot can reach to 100%, so the device need tune SAR based on 100% UL slots in one frame. But we cannot always configure all the slot to UL in long time, in order to tune SAR more convenient, device can use fixed UL slot configure, the calculate the target SAR setting based on the UL slot duty cycle.

For none HPUE band, the max Tx power have no change when UL configure change

- e.g. n40 use the 40% UL slot to test the SAR and get the SAR limitation x dBm, consider 100% UL slot is the worst case, the target setting value should be around $x - 10 \cdot \log_{10}(1/0.4)$ dBm.< the final setting should be refer to the final test in the lab)

For HPUE band, the SAR setting is quite special, based on the 3Gpp, when the UL slot duty cycle higher than 50%, the max Tx power should back off 3dB, it means the average Tx power (PC2=26dBm) in 50% duty cycle is similar with 100% (23dBm), so the calculate solution should be refer to 50% duty cycle:

- e.g. n41 use the 40% UL slot w/ HPUE to test SAR and get the SAR limitation x dBm, consider 100% UL slot is the worst case, the target setting value should be around $x - 10 \cdot \log_{10}(0.5/0.4)$ dBm. < the final setting should be referred to the final test in the lab>.

4.3.7.2 For LTE

For none HPUE band ,3Gpp defined the max UL slots is 6, so the device need tune SAR based on 60% UL slots in one frame.

For HPUE band, similar as NR band, when UL slots higher (or equal) than 50%, the power will back off 3dB, it means the average power w/ 4 UL slots (PC2=26dBm) will be higher than the power w/ 6 UL slots(23dBm), so should use the 4 UL slots to do the SAR turning.

NOTES: When do the SAR tuning and achieve the target SAR value, we need consider the Tx max power tolerance (tune up power)

4.3.8 Appendix

4.3.8.1 Band Bitmap

4.3.8.1.1 WCDMA

Table 2. WCDMA Bands

Band Bitmap	Band Name
"0x1"	WCDMA_I_IMT_2000
"0x2"	WCDMA_II_PCS_1900
"0x8"	WCDMA_IV_1700
"0x10"	WCDMA_V_850
"0x80"	WCDMA_VIII_900

4.3.8.1.2 LTE

Table 3. LTE Bands

Band Bitmap	Band Name
"0x1 "	BAND1
"0x2"	BAND2
"0x4"	BAND3

Band Bitmap	Band Name
"0x8"	BAND4
"0x10"	BAND5
"0x40"	BAND7
"0x80"	BAND8
"0x800"	BAND12
"0x1000"	BAND13
"0x2000"	BAND14
"0x10000"	BAND17
"0x20000"	BAND18
"0x40000"	BAND19
"0x80000"	BAND20
"0x1000000"	BAND25
"0x2000000"	BAND26
"0x8000000"	BAND28
"0x10000000"	BAND29
"0x20000000"	BAND30
"0x80000000"	BAND32
"0x200000000"	BAND34
"0x2000000000"	BAND38
"0x4000000000"	BAND39
"0x8000000000"	BAND40
"0x10000000000"	BAND41
"0x20000000000"	BAND42
"0x40000000000"	BAND43
"0x200000000000"	BAND46
"0x800000000000"	BAND48
"0x2000000000000000"	BAND66
"0x4000000000000000"	BAND71

4.3.8.1.3 5G NR

Table 4. 5G NR Bands

Band Bitmap	Band Name
"0x1"	N1
"0x2"	N2
"0x4"	N3
"0x10"	N5
"0x40"	N7
"0x80"	N8
"0x80000"	N20
"0x1000000"	N25
"0x8000000"	N28
"0x20000000"	N30
"0x2000000000"	N38
"0x8000000000"	N40
"0x10000000000"	N41
"0x800000000000"	N48
"0x200000000000000000"	N66
"0x4000000000000000000"	N71
"0x10000000000000000000"	N77
"0x20000000000000000000"	N78
"0x40000000000000000000"	N79
"0x80000000000000000000"	N80
"0x100000000000000000000"	N81
"0x200000000000000000000"	N82
"0x400000000000000000000"	N83
"0x800000000000000000000"	N84

4.3.8.1.4 ENDC

Table 5. ENDC Bands

[illegible]

[illegible]

"0x800000"	BAND28	"0x400000000000000000000000"	N83
"0x200000000"	BAND30	"0x10"	N5
"0x20000000000"	BAND38	"0x200000000000000000000000"	N78
"0x40000000000"	BAND39	"0x100000000000"	N41
"0x40000000000"	BAND39	"0x400000000000000000000000"	N79
"0x80000000000"	BAND40	"0x100000000000"	N41
"0x80000000000"	BAND40	"0x100000000000000000000000"	N77
"0x100000000000"	BAND41	"0x100000000000"	N41
"0x100000000000"	BAND41	"0x100000000000000000000000"	N77
"0x8000000000000000"	BAND48	"0x10"	N5
"0x8000000000000000"	BAND48	"0x200000000000000000000000"	N66
"0x200000000000000000000000"	BAND66	"0x10"	N5
"0x200000000000000000000000"	BAND66	"0x400000000000000000000000"	N71

New add ENDC bands for RF requirement (2021-01-20), as below.

Table 6. ENDC Bands in RF Requirement

Band Value	Band Name	Band Value	Band Name
"0x02"	BAND2	"0x8000000000000000"	N48
"0x02"	BAND2	"0x100000000000000000000000"	N77
"0x800000000000"	BAND5	"0x100000000000000000000000"	N77
"0x800"	BAND12	"0x100000000000000000000000"	N77
"0x1000"	BAND13	"0x100000000000000000000000"	N77
"0x2000"	BAND14	"0x100000000000000000000000"	N77
"0x200000000"	BAND30	"0x100000000000000000000000"	N77
"0x200000000000000000000000"	BAND66	"0x8000000000000000"	N48
"0x200000000000000000000000"	BAND66	"0x100000000000000000000000"	N77
...	...	...	...

4.4 Thermal

1. Show temperature of module sensors and Connect Band, Power, RAT, Thermal Throttling ON and OFF.

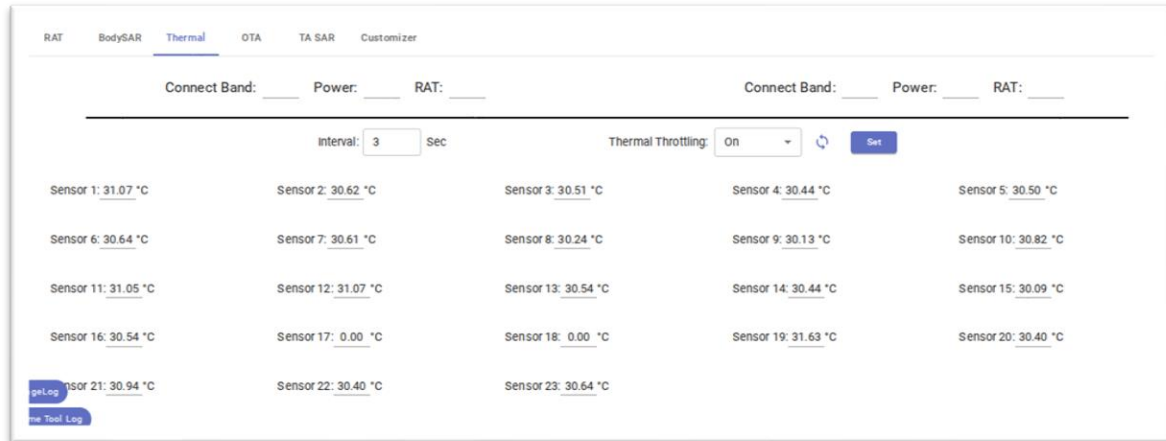


Figure 4-25. Thermal Display

2. Select the drop-down box and click **Set** button to ON or OFF Thermal Throttling.
3. Thermal provides the ability to customize automatic refresh intervals, which are 3 seconds by default.
4. When the Thermal function is enabled, it will automatically record the temperature of Sensors and save the data in the folder located under the Engineer Tool directory. The file is saved in **.csv** format and named by date.

4.5 OTA

This unit is used by Rolling customers for OTA debugging.

4.5.1 The Main Interface

4.5.1.1 Supported Functions

This unit supported functions are as follows:

To switch antenna RX function.

4.5.1.2 Main Interface

The main interface is as follows:

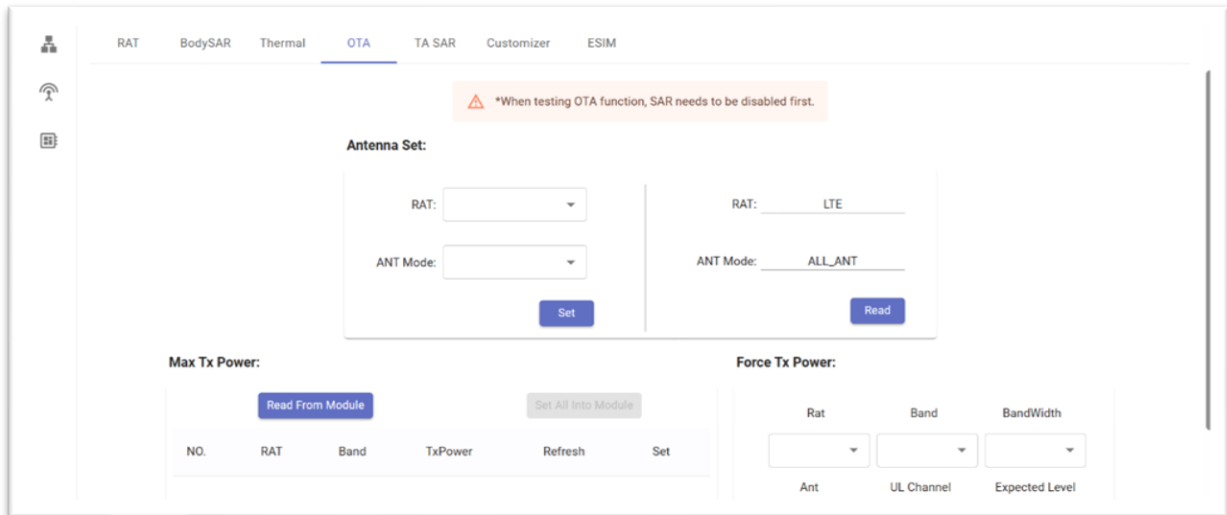


Figure 4-26. Main Interface of MTK platform

4.5.2 Antenna

4.5.2.1 Configure Antennas

4.5.2.1.1 Select RAT

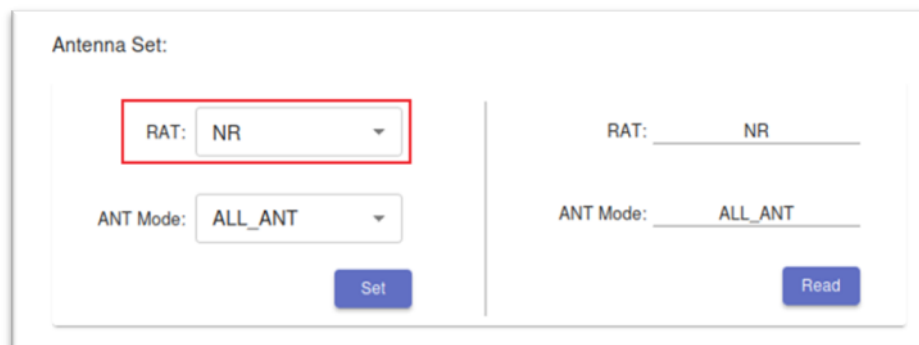


Figure 4-27. Select RAT of MTK platform



Figure 4-28. Select RAT of Qualcomm platform

This area is to select the enable RAT.

Options:

- UMTS (MTK platform only)
- LTE
- NR-RAN

4.5.2.1.2 Set ANT Mode

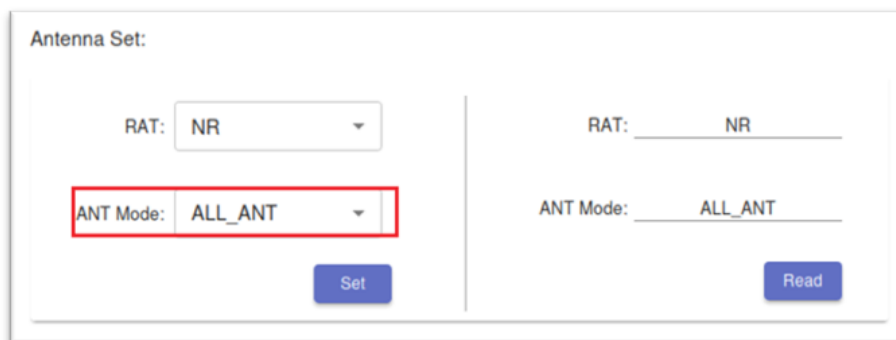


Figure 4-29. Set ANT Mode of MTK platform



Figure 4-30. Set ANT Mode of Qualcomm platform

This area is to select the antenna mode.

For module of Qualcomm platform, it has only two options as follows:

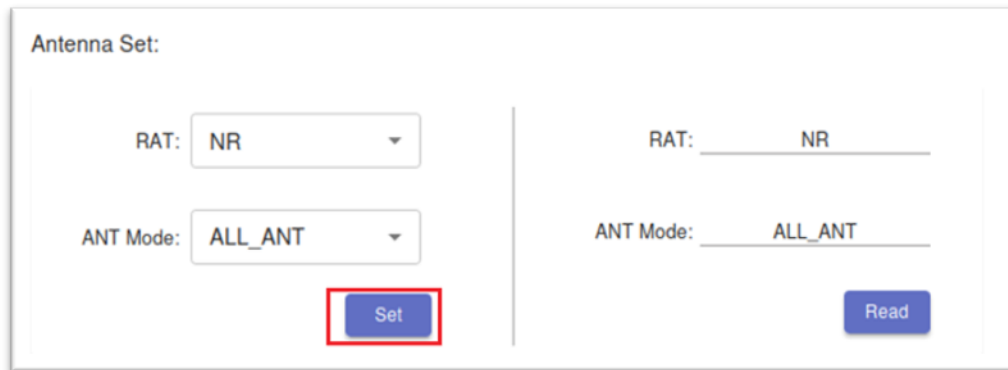
- ANT0
- ANT1.

For module of MTK platform, it has five options as follows:

- Main Only
- Aux Only
- MIMO Main Only
- MIMO Aux Only

- ALL_ANT

4.5.2.1.3 Set to Module



Antenna Set:

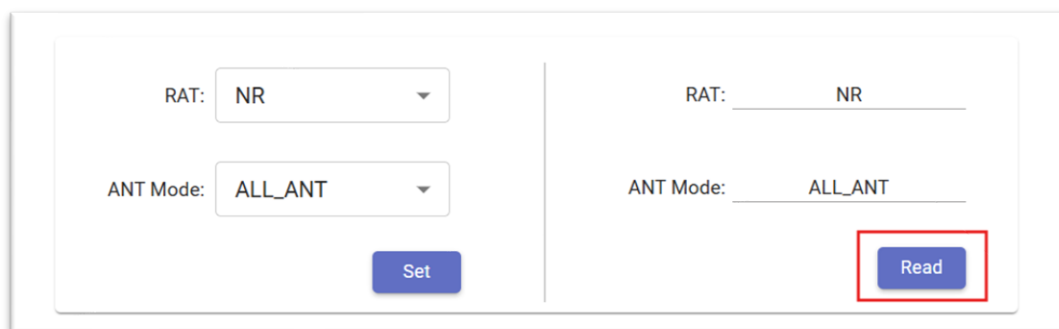
RAT: NR	RAT: NR
ANT Mode: ALL_ANT	ANT Mode: ALL_ANT
Set	Read

Figure 4-31. Set Ant to Module

Click the **Set** button, the antenna setting you select will configure to module.

4.5.2.2 Read the Configuration

Click the **Read** button, the OTA unit will read the antenna configurations from module and show it.



Antenna Set:

RAT: NR	RAT: NR
ANT Mode: ALL_ANT	ANT Mode: ALL_ANT
Set	Read

Figure 4-32. Read Ant from module

4.5.3 Max Tx Power

4.5.3.1 Read & Set

Max Tx Power:

Read From Module
Set All Into Module

NO.	RAT	Band	TxPower	Refresh	Set
1	UMTS	BAND 1	188	Refresh	Set
2	UMTS	BAND 2	188	Refresh	Set
3	UMTS	BAND 4	188	Refresh	Set
4	UMTS	BAND 5	188	Refresh	Set
5	UMTS	BAND 8	188	Refresh	Set
6	LTE	BAND 1	184	Refresh	Set

Figure 4-33. Max TX Power Display

1. Click **Read from Module** button to obtain the current Max Tx Power value of each Band of the module.
2. Click **Set All into Module** button to set the Tx power value of each Band in the interface to the module. After the setting is successful, the value of Tx Power on the interface will change to the value of the new setting.
3. Click **Set** button to set single band's Tx Power.
4. Click **Refresh** button to sync single band's Tx Power with modem.

4.5.3.1.1 Max Tx Power Value Range

For Module RW350R:

- UMTS value: 0-200. Example: $200 = 200/8 = 25\text{dBm}$
- LTE value: 0-232. Example: $232 = 232/8 = 29\text{dBm}$
- NR-RAN value: 0-232. Example: $232 = 232/8 = 29\text{dBm}$

For Module RW135/RW135R:

- UMTS value: 0-200. Example: $200 = 200/10 = 20\text{dBm}$
- LTE value: 0-232. Example: $230 = 230/10 = 23\text{dBm}$

- NR-RAN value: 0-232. Example: $230 = 230/10 = 23\text{dBm}$



- Max Tx Power needs to be reset when testing TASAR or Body SAR. Failure to do so will affect testing of TASAR functionality.
- This function only supports power adjustment of a single band, not ENDC power adjustment.

Known issue: The max Tx Power of LTE and NR-RAN may not take effect when the value is successfully set to 216 to 232 by AT.

4.5.3.1.2 Table Introductions

- Column 1 of the table data indicates the index of the Max Tx Power setting item.
- Column 2 of the table data indicates the RATs of mobile communication technology, include UMTS, LTE, and NR-RAN.
- Column 3 of the table data indicates the bands of different RATs.
- Column 4 of the table data indicates the Default Tx Power values of different bands.

4.5.3.1.3 Function Introduction

- Click the **Read From Module** to read the Max Tx Power data from the module.
- Click the **Set All Into Module** set all Max Tx Power data into module. It takes a few seconds to wait for completion.
- Input the value in .
- Click **Set** to set one Max Tx Power data into module.

4.5.4 Force Tx Power

The function of this unit is to test whether the Tx Power value can achieve the setting in this window.



Only professional personnel appointed by customer could use this unit, others may cause heat release damage to the module due to improper use.

- Connect module to test instrument.
 - Select RAT
- If you select "WCDMA", it needs enter Channel and select Band, ANT, and BW.

Rat	Band	BandWidth	Ant	UL Channel						
WCDMA	1	5MHZ	M2							
<table border="1"> <thead> <tr> <th>Expected Level</th> <th>Power</th> <th>MCS</th> </tr> </thead> <tbody> <tr> <td>-95</td> <td>24</td> <td>QPSK</td> </tr> </tbody> </table>					Expected Level	Power	MCS	-95	24	QPSK
Expected Level	Power	MCS								
-95	24	QPSK								
ON TxPower OFF TxPower										

Figure 4-34. Force Tx Power Select WCDMA

- If you select "LTE", it needs enter Frequency(KHz) and select Band, ANT, BW, MCS, 1st RB, and RBs.

Rat	Band	BandWidth	Ant	UL Channel										
LTE	B18	5MHZ	M											
<table border="1"> <thead> <tr> <th>Expected Level</th> <th>Power</th> <th>MCS</th> <th>RB Start</th> <th>RB Num</th> </tr> </thead> <tbody> <tr> <td>-95</td> <td>24</td> <td>QPSK</td> <td>0</td> <td>25</td> </tr> </tbody> </table>					Expected Level	Power	MCS	RB Start	RB Num	-95	24	QPSK	0	25
Expected Level	Power	MCS	RB Start	RB Num										
-95	24	QPSK	0	25										
ON TxPower OFF TxPower														

Figure 4-35. Force Tx Power Select LTE

- If you select "NR-RAN", it needs enter Frequency(KHz) and select Band, SCS, ANT, BW, MCS, 1st RB, and RBs.

Rat	Band	BandWidth	Ant	UL Channel												
NR-RAN	N38	10MHZ	M2													
<table border="1"> <thead> <tr> <th>Expected Level</th> <th>Power</th> <th>MCS</th> <th>RB Start</th> <th>RB Num</th> <th>NR SCS</th> </tr> </thead> <tbody> <tr> <td>-95</td> <td>24</td> <td>DFT_S_BPSK</td> <td>0</td> <td>50</td> <td>15KHZ</td> </tr> </tbody> </table>					Expected Level	Power	MCS	RB Start	RB Num	NR SCS	-95	24	DFT_S_BPSK	0	50	15KHZ
Expected Level	Power	MCS	RB Start	RB Num	NR SCS											
-95	24	DFT_S_BPSK	0	50	15KHZ											
ON TxPower OFF TxPower																

Figure 4-36. Force Tx Power Select NR-RAN

1. Enter Power. Power value from 0 to 26 for RW135R and RW350R, and from 0 to 24 for RW135. Its' unit is dBm.
2. Click **ON Tx Power** to set information of left side to module, then module will transmit the power value on this antenna.
3. Compare the set power in the windows and the measurement on instrument.
4. If tests complete, to Click **OFF Tx Power**.



1. The 3GPP protocol states that when a Band is registered, it is required to register the central frequency. Low/High value indicates the illegal frequency. For details, please refer to the 3GPP protocol for setting parameters of each RAT and Band's frequency or channel value and others.

3GPP protocol URL: <https://www.3gpp.org/>

2. When RAT chooses "NR-RAN", Frequency input should meet the following requirements:

0~3000 MHz: the selected frequency point must be an integer multiple of 5KHz;

3000~24250Mhz: the selected frequency point must be an integer multiple of 15 KHz.

3. Do not insert a SIM card during the test to avoid AT delivery failure. Because of when SIM card is inserted, AT may fail to be delivered, which is the expected logic of MTK AT instruction non-signaling test.

4. In N41/N77/N78/N79, in addition to the switch between M and M2 port, the switch of other antenna ports needs Engineer Tool automatically sends the command to restart the module.

5. On the RW350R platform, forced transmit power control on the M port for LTE Band 41 is not supported because of inherent platform limitations.

4.6 TA SAR

The TA SAR Tuning unit is used by Rolling customers for TA SAR debugging. In this unit, the user can easily configure the TA SAR table of the module. OTA—>MAX Tx Power needs to be reset when testing TASAR.

Note: Prior to executing these testing functions on Windows or Linux systems, please ensure that the **Service Flash** option is disabled from the **Telephony Tool Download** page.

4.6.1 The Main Interface

4.6.1.1 Supported Functions

This unit supported functions are as follows:

- TA SAR switch
- TA SAR table configuration

- TA SAR Quick Set

4.6.1.2 Main Interface

The main interface is as follows:

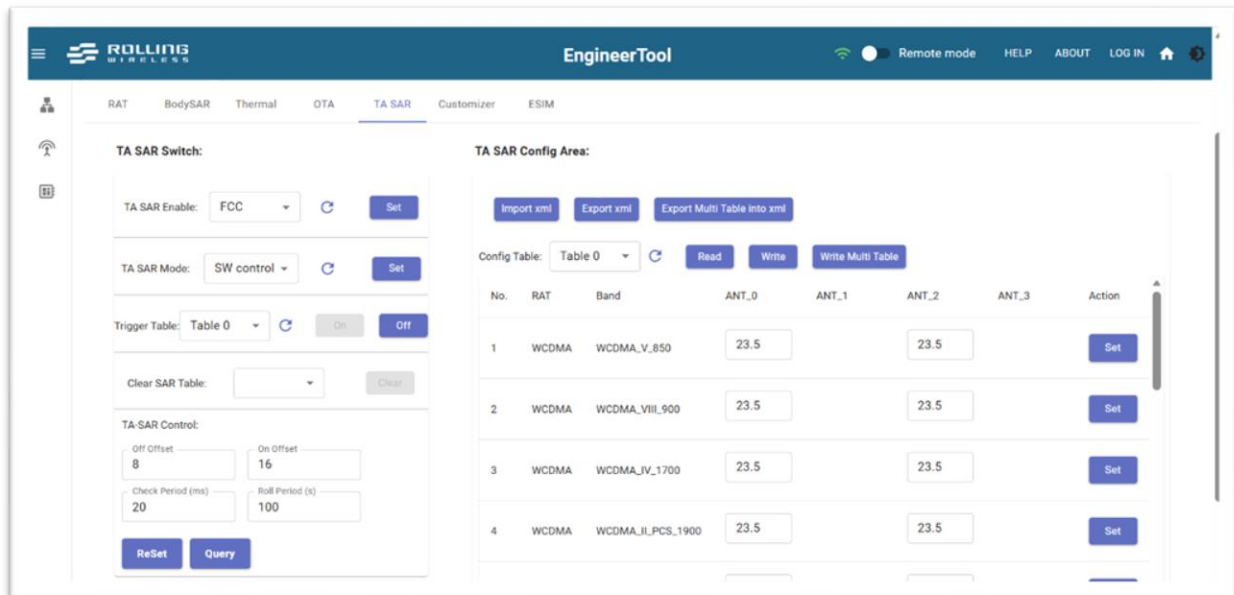


Figure 4-37. Main Interface

4.6.2 TA SAR Switch

4.6.2.1 TA SAR Enable

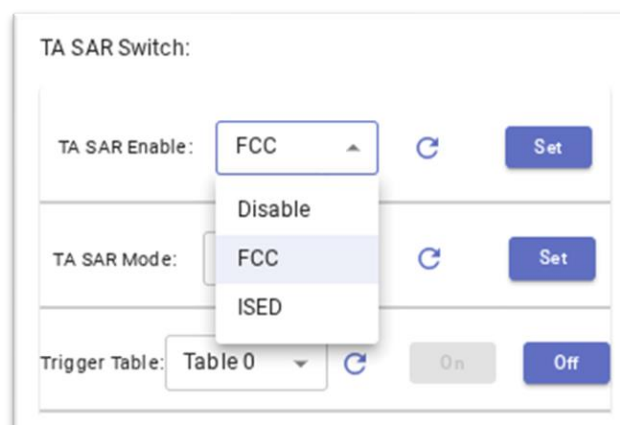


Figure 4-38. TA SAR Enable

This area is the enable switch of the TA SAR function. When the setting is Enables, the entire UI can be operated; when the setting is Disable, all items are inoperable except for the TA SAR Enable.

1. The Set TA SAR Enable drop-down box is used to display the current TA SAR Mode Enable status. Click the **refresh** button on the right to refresh the status.
2. Click the **Set** Button to set FCC TA SAR or ISED TA SAR to Enable or Disable. After clicking, the current Enable status of the module will be automatically read and displayed. If it cannot be modified, the setting is unsuccessful.

4.6.2.2 TA SAR Mode

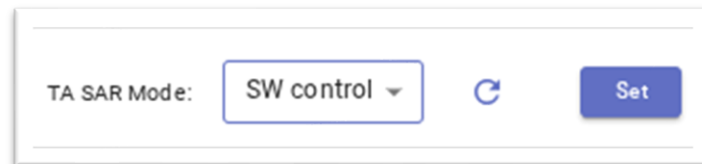


Figure 4-39. TA SAR Mode Switch

This area is to set the control mode of TA SAR, including hardware control (HW control) and software control (SW control):

1. Set TA SAR Mode drop-down box is used to display the current control mode. You can click the **refresh** button on the right to read the value from the module.
2. Click the **Set** Button to set the TA SAR Mode to HW control or SW control. After clicking, the current mode of the module will be automatically read and displayed. If it cannot be modified, the setting is unsuccessful.

4.6.2.3 Trigger Table

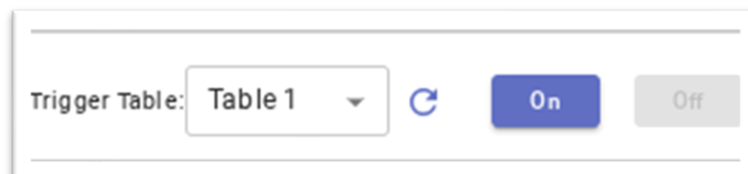


Figure 4-40. Trigger Table Switch

This area is to set the currently used Table and to enable or disable the trigger table.

1. The **Set Trigger Table** drop-down box will display the currently used Table. Click the refresh button on the right to read the value from the module.
2. Click the **Set Trigger Table** drop-down box to set the TA SAR Table, the current Table of the module will be automatically read and displayed. If it cannot be modified, the setting is unsuccessful.
3. Click the **ON** Button to enable the trigger table.
4. Click the **OFF** Button to disable the trigger table.

4.6.2.4 Clear SAR Table

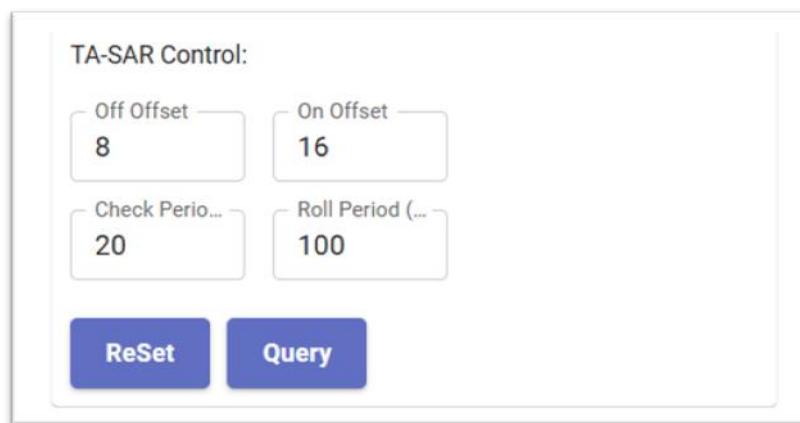


Clear SAR Table:

Figure 4-41. Clear SAR Table

This area is used to clear the TA SAR value. If the click **Clear**, the antenna will work at the maximum power.

4.6.2.5 TA SAR Control



TA-SAR Control:

Off Offset 8	On Offset 16
Check Perio... 20	Roll Period (...) 100
ReSet	Query

Figure 4-42. TA SAR Control

Off Offset:

(0-184) dpr off offset, in 1/8dB unit. For calculate the lower threshold (8 is recommended), off offset always less than on offset, off offset < on offset. Default value is 0.

On Offset:

(0-184) dpr on offset, in 1/8dB unit. For calculate the dpr on TX power (FCC:16 is recommended; ISED: 32 is recommended), off offset is always less than on offset, off offset < on offset. Default value is 0.

Check Period:

(20-1000) TAS check period (recommend 1000), in 1 millisecond unit. Check period is always less than roll period times 1000, check period < roll period*1000. Default value is 0.

Roll Period:

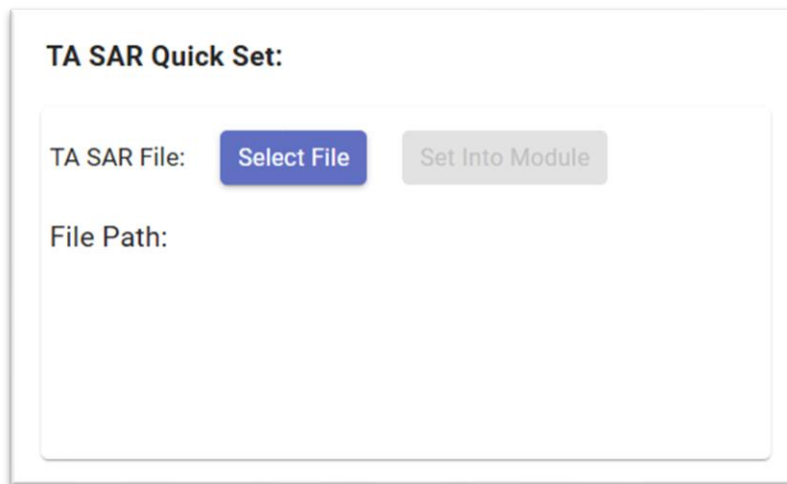
(0-65533) TAS roll period (recommend 100), in 1 second unit. Check period(ms) is always less than roll_period*1000(ms), check period(ms) < roll_period*1000(ms). Default value is 0.

Click **Reset** Button to reset the TAS Control parameter corresponding to the current SAR mode.

Click **Query** Button to query the TAS Control parameter corresponding to the current SAR Mode.

4.6.3 TA SAR Quick Set

TA SAR Quick Set sets the maximum transmit power corresponding to all mobile communication technologies (3G/4G/5G) and frequency bands supported by the module one by one by reading the xml file; if the setting item is not supported by the module, the setting fails. The content of a single file includes tech (mobile communication technology), band (frequency band) and power (maximum transmit power).



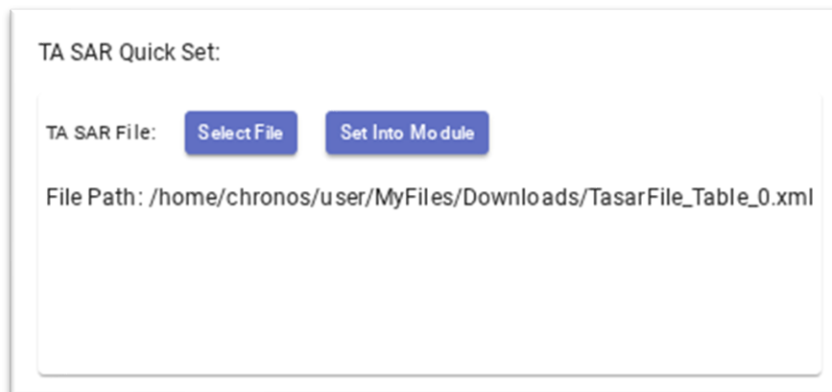
TA SAR Quick Set:

TA SAR File:

File Path:

Figure 4-43. TA SAR Quick Set

1. Click **Select File** to open the file selection dialog box. The required XML document can be selected from dialog box.



TA SAR Quick Set:

TA SAR File:

File Path: /home/chronos/user/MyFiles/Downloads/TasarFile_Table_0.xml

Figure 4-44. After selecting the file

2. After selecting the file, it will check whether the content of the file is valid. If it is valid, the full path of the file will be displayed in the text, as shown below.
3. Click the **Set Into Module** to set the content of the file to the module line by line. During the setting process, the entire UI is grayed out and cannot be operated. The UI will not return to normal until all data is set or the setting fails.

4.6.4 TA SAR Table Configuration

4.6.4.1 Export Multiple Tables



TA SAR Config Area:

Buttons: Import xml, Export xml, **Export Multi Table into xml** (highlighted with a red box)

Config Table: Table 0 (dropdown) [Refresh icon] [Read] [Write] [Write Multi Table]

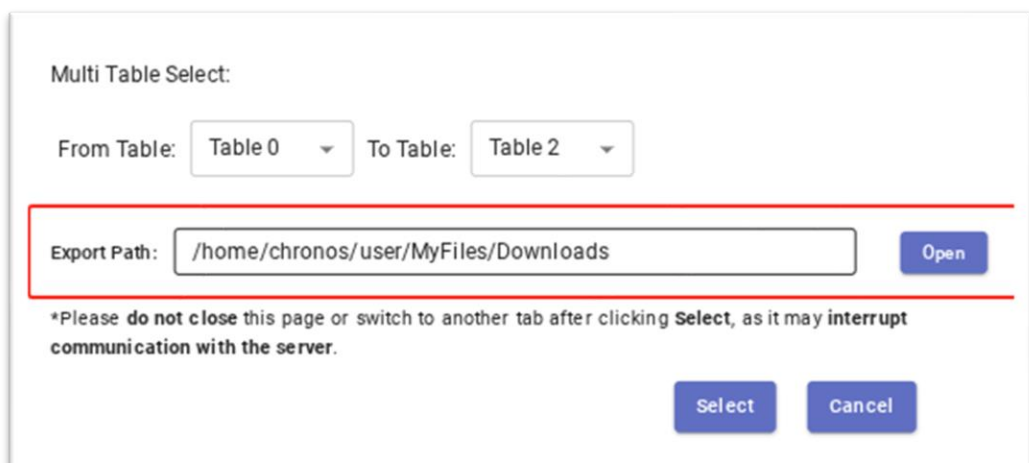
No.	RAT	Band	ANT_0	ANT_1	ANT_2	ANT_3	Action
1	WCDMA	WCDMA_V_850	23.5		23.5		[Set]
2	WCDMA	WCDMA_VIII_900	23.5		23.5		[Set]

Figure 4-45. Export Multi Table into xml

For example, click **Export Multi Table into xml**, you can get a pop-up and choose a range of tables that you want to export.

Click **Open** to open the export folder for easy browsing of the exported xml file. The export path is fixed to the TA SAR folder under the application path.

1. FCC mode the path is TA_sar, ISED mode the path is ISED_TA_sar.

Multi Table Select:

From Table: Table 0 (dropdown) To Table: Table 2 (dropdown)

Export Path: /home/chronos/user/MyFiles/Downloads [Open]

***Please do not close this page or switch to another tab after clicking **Select**, as it may interrupt communication with the server.**

[Select] [Cancel]

Figure 4-46. Export Path

2. Before performing FCC TASAR, enable FCC TASAR first; Similarly ,before performing

ISED TASAR, enable ISED TASAR first

4.6.4.2 Config Table

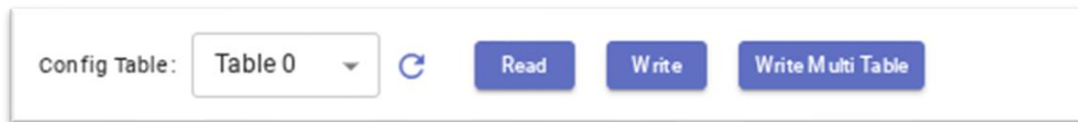


Figure 4-47. Config Table Switch

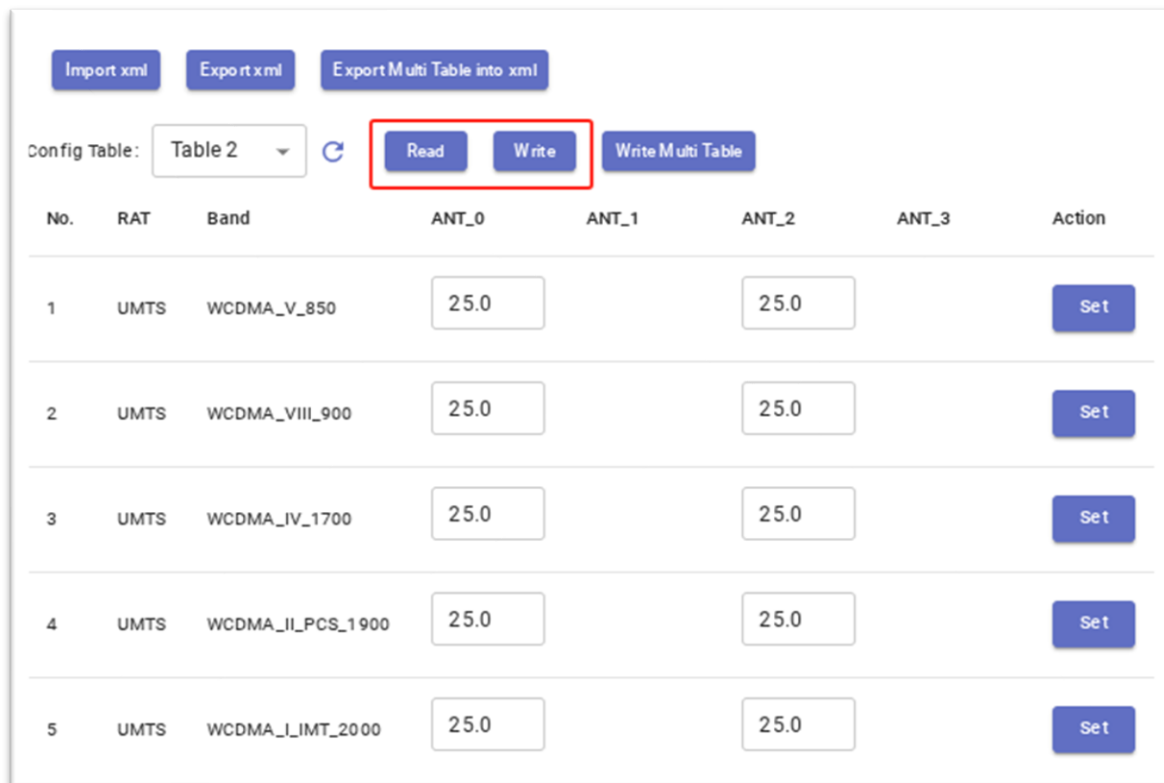
1. This area is used to configure the TA SAR Table that currently needs to be modified.
2. **Config Table** is to display the table currently being modified. Click the refresh button on the right to read the value from module.



: Refresh the selected Config Table.

3. Click the **Config Table** drop-down box to set the TA SAR Mode, and click the **read** button to set. After reading, it will automatically read the current configuration table from the module and display it on the interface.

4.6.4.3 Read & Write TA SAR



No.	RAT	Band	ANT_0	ANT_1	ANT_2	ANT_3	Action
1	UMTS	WCDMA_V_850	25.0		25.0		Set
2	UMTS	WCDMA_VIII_900	25.0		25.0		Set
3	UMTS	WCDMA_IV_1700	25.0		25.0		Set
4	UMTS	WCDMA_II_PCS_1900	25.0		25.0		Set
5	UMTS	WCDMA_I_IMT_2000	25.0		25.0		Set

Figure 4-48. Read & Write TA SAR

This item can import table, export table, edit table, read and set table.

UI introduction:

1. Column 1 of the table data indicates the Index of the TA SAR setting item.
2. Column 2 of the table data indicates the type of mobile communication technology.
3. Column 3 of the table data indicates the frequency band value
4. Column 4 to Column 7 of the table data indicate the power values of different antennas (unit: 1/8 dB);

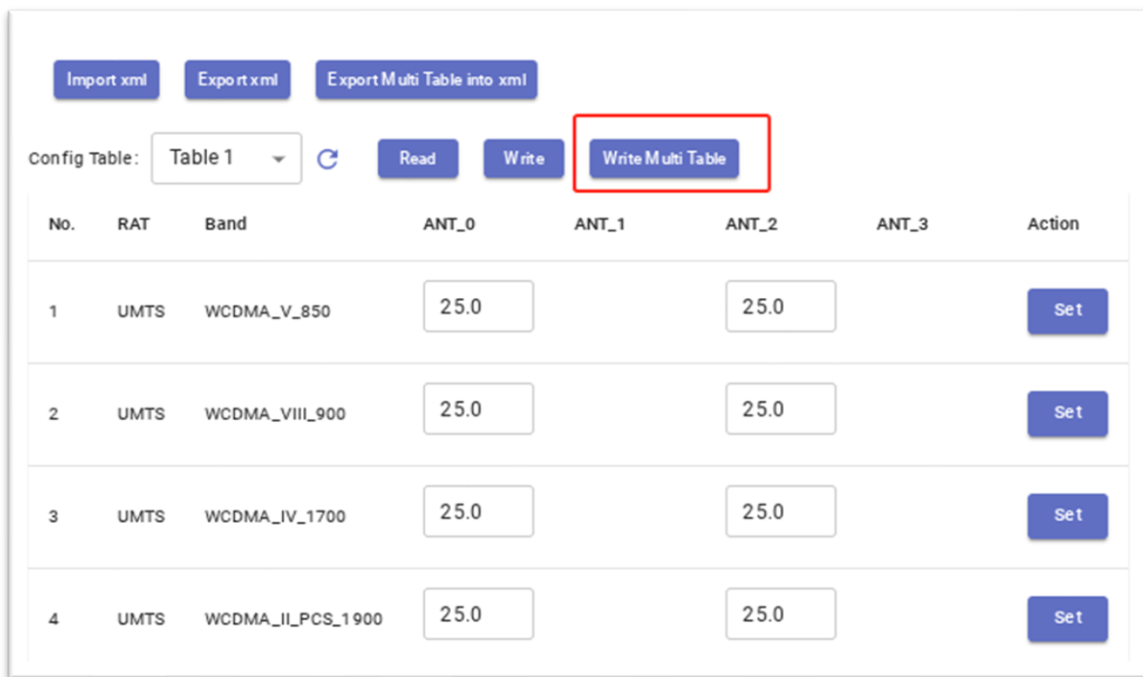
Function introduction:

1. Click the **Set** behind each row to set the data in the corresponding row of the module as row-change data, which is a single row modification.
2. Click the **Read** to read the TA SAR data of the current table from the module.
3. Click the **Write** to synchronously set the current table data row by row to the corresponding row of the module. If there is a row in the middle of the setting fails, all subsequent settings will be aborted. After clicking the Set Into Module button, the entire interface will be inoperable until the setting is completed or the setting is abnormally terminated, and the entire UI returns to normal.
4. Click **Import** to import an xml file to the table.
5. Click **Export** to export the table data to an xml file; the file path is fixed to the TA SAR folder under the application path, and the export file naming rule is TA SARFile_time.xml.



1. In ISED mode, the value showed on the UI will be 12 higher than the real value. When clicking **Set** or **Write**, the value set will be 12 lower than the value showed on the UI.
2. In ISED mode, when clicking **Export**, the value export will be 12 lower than the value showed on the UI, and the file naming rule is TA_SAR_ISED_File_Table_X.xml.
3. The power value displayed in the tool interface is increased by 1.5dBm compared to the value displayed in the NV.

4.6.4.4 Write Multi Table



Import xml Export xml Export Multi Table into xml

Config Table: Table 1

No.	RAT	Band	ANT_0	ANT_1	ANT_2	ANT_3	Action
1	UMTS	WCDMA_V_850	25.0		25.0		Set
2	UMTS	WCDMA_VIII_900	25.0		25.0		Set
3	UMTS	WCDMA_IV_1700	25.0		25.0		Set
4	UMTS	WCDMA_II_PCS_1900	25.0		25.0		Set

Figure 4-49. Write Multi Table

1. Click the **"Write Multi Table"** button to write TASAR value.
2. Select the TASAR tables you want to write.



Multi Table Select:

From Table: Table 0 To Table: Table 10

*Please do not close this page or switch to another tab after clicking **Select**, as it may interrupt communication with the server.

Figure 4-50. Write data to all selected tables

3. Click "Select" in the pop-up window to confirm.

4.6.5 SAR Up threshold setting for TDD band

The SAR value setting is similar with Body SAR, the only difference is:

For LTE TA-SAR setting is the average power Up limit for SAR based on all time cycle, if we want to calculate the TA-SAR average power up limit based on above Body SAR value, we need calculate the real average power.

e.g B40 testing and tuning UL slots is 6, and use the target SAR power y dBm, because the y dBm is the max Tx power, the calculated average power should be: $y + 10 \cdot \log_{10}(6/10)$ dBm.

So the theoretical TA-SAR average power up limit should be: $y + 10 \cdot \log_{10}(6/10)$ dBm. .<the final setting should be refer to the final test in the lab>.

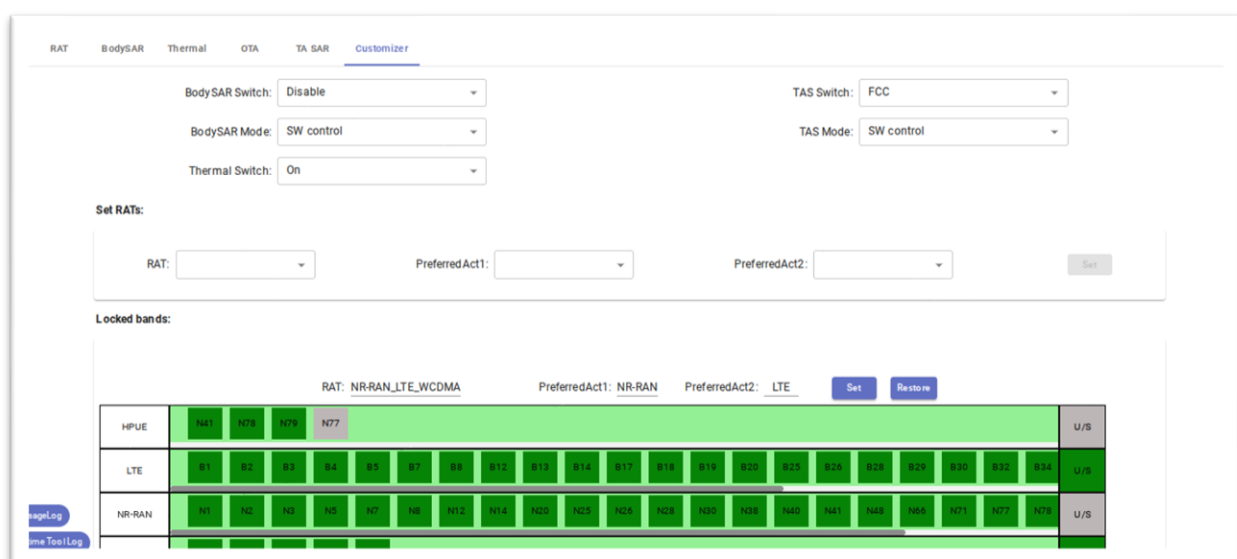
For NR, if want to calculate the target value based on above body SAR, we can use the body SAR value directly, because the NR TDD band max UL slot is considered to 100%, and above body SAR have already considered the time slot impact.

NOTES: When do the SAR tuning and achieve the target SAR value, we need consider the Tx max power tolerance (tune up power)

4.7 Customizer

The function of Customizer is exporting the WwanFunctionConfigureFile.xml based on selected on UI.

WwanFunctionConfigureFile.xml is a file that include WWAN features disable and enable customization information, the customization information as shown in the [Functions list](#). This customization page can take effect after flashing the information of WwanFunctionConfigureFile.xml to modem. The usage method of WwanFunctionConfigureFile.xml is in the webserver user guide.



The Customizer interface includes the following sections:

- Navigation Tabs:** RAT, BodySAR, Thermal, OTA, TA SAR, Customizer (selected).
- BodySAR Settings:**
 - BodySAR Switch: Disable
 - BodySAR Mode: SW control
 - Thermal Switch: On
- TAS Settings:**
 - TAS Switch: FCC
 - TAS Mode: SW control
- Set RATs:**
 - RAT: [Dropdown]
 - PreferredAct1: [Dropdown]
 - PreferredAct2: [Dropdown]
 - [Set Button]
- Locked bands:**
 - RAT: NR-RAN_LTE_WCDMA
 - PreferredAct1: NR-RAN
 - PreferredAct2: LTE
 - [Set Button] [Restore Button]

HPUE	N41	N78	N79	N77																	U/S	
LTE	B1	B2	B3	B4	B5	B7	B8	B12	B13	B14	B17	B18	B19	B20	B25	B26	B28	B29	B30	B32	B34	U/S
NR-RAN	N1	N2	N3	N5	N7	N8	N12	N14	N20	N25	N26	N28	N30	N38	N40	N41	N48	N66	N71	N77	N79	U/S

Figure 4-51. Customization Main interface

4.7.1 Functions list

Table 7. Description of feature customization options

Feature Name	Customization Item	Note
Bands Enable	Bands supported	Disable or Enable bands in Bands Enable
Body SAR Switch	Disable	Disable Body SAR
	Enable	If Body SAR Switch is Enable , TAS Switch will be Disable . If TAS Switch is Enable , Body SAR Switch will be Disable .
Body SAR Mode	SW	Software Mode
	HW	Hardware Mode
RATs	UMTS	Locked RAT is UMTS.
	LTE	Locked RAT is LTE.
	LTE/UMTS	Locked RATs are LTE and UMTS.
	NR-RAN	Locked RAT is LTE and UMTS.
	NR-RAN/WCDMA	Locked RATs are NR-RAN and WCDMA.
	NR-RAN/LTE	Locked RATs are NR-RAN and LTE.
	NR-RAN/LTE/WCDMA	Locked RATs is NR-RAN and LTE and WCDMA.
RAT_Prefer_1	UMTS	First priority RAT is UMTS.
	LTE	First priority RAT is LTE.
	NR-RAN	First priority RAT is NR-RAN.
RAT_Prefer_2	UMTS	Second priority RAT is UMTS.
	LTE	Second priority RAT is LTE.

Feature Name	Customization Item	Note
	NR-RAN	Second priority RAT is NR-RAN.
Thermal Switch	Disable	Disable thermal
	Enable	Enable thermal
TAS Switch	Disable	Disable TA-SAR.
	Enable	Enable TA-SAR.
TAS Mode	SW	Software Mode
	HW	Hardware Mode

4.7.2 Operation Steps

BodySAR Switch: Disable

BodySAR Mode: SW control

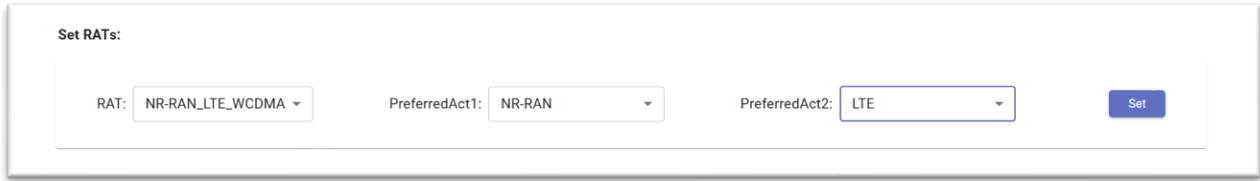
Figure 4-52. Body Sar Parameter

LTE	B1	B2	B3	B4	B5	B7	B8	B12	B13	B14	B17	B18	B19	B20	B25	B26	B28	B29	B30	B32	B34	B36	B38	B39	B40	U/S
NR-RAN	N1	N2	N3	N5	N7	N8	N20	N25	N28	N30	N38	N41	N66	N71	N77	N78	N79									U/S
WCDMA	1	2	4	5	6																					U/S

Figure 4-53. Band Enable Display

1. Click  to expand drop-down box.
2. Select function customization items from drop-down box.
3. Click button like **1** to disable or enable this band, and click button **U/S** to disable or enable all bands of this RAT. Gray bands will be disabled; green bands will be enabled.
4. Click **Export** to generate the WwanFunctionConfigureFile.xml.
5. Click **Open Folder** to access the path to save WwanFunctionConfigureFile.xml.

4.7.3 Set RATs



Set RATs:

RAT: NR-RAN_LTE_WCDMA PreferredAct1: NR-RAN PreferredAct2: LTE Set

Figure 4-54. RAT Preferred

1. Configure RATs information to provide configurable items that are supported by the current module (obtained from the module side), including RAT, PreferredAct1, PreferredAct2.
2. RAT, PreferredAct1, PreferredAct2 are available from supported pull-down options.
 - RAT: Radio Access Technology with UMTS (3G), LTE (4G), NR-RAN (5G), Automatic (automatic), LTE/UMTS (3G/4G dual mode), NR-RAN/WCDMA (3G/5G dual mode), NR-RAN/LTE (4G/5G dual mode), NR-RAN/WCDMA/LTE (3G/4G/5G triple mode) and more.
 - PreferredAct1&PreferredAct2: First and second preferably, there are WCDMA (3G), LTE (4G), NR-RAN (5G) and other options.
3. PreferredAct1 and PreferredAct2 are not operational when the RAT selection is a single item (UMTS, LTE, NR-RAN).
4. When the RAT selection is a two-item (LTE/UMTS, NR-RAN/WCDMA, NR-RAN/LTE), PreferredAct1 is operational, and what appears in its pull-down box is a split item of the RAT double-item;
5. When the RAT selection is three-item (NR-RAN/WCDMA/LTE), both PreferredAct1 and PreferredAct2 can be operated, with the content shown in the PreferredAct1 pull-down box being a split item for the RAT double item, and the content shown in the PreferredAct2 pull-down box showing two other items in the RAT double-item split item except the PreferredAct1 selection;
6. When the RAT selection is Automatic, neither PreferredAct1 and PreferredAct2 are operational, and their set-up functions restore the default values for all bands on the module side, the same effect as Restore in Set bands;
7. When the format is set, the band of the format is locked to all bands it supports.

4.7.4 Locked Bands

1. When bands are set, their corresponding format is the module's current format, as shown in the red box below:

Locked bands:

RAT: NR-RAN_LTE_WCDMA PreferredAct1: NR-RAN PreferredAct2: LTE

Set **Restore**

HPUE	N41	N77	N78	N79																	U/S
LTE	B1	B2	B3	B4	B5	B7	B8	B12	B13	B14	B17	B18	B19	B20	B25	B26	B28	B29	U/S		
NR-RAN	N1	N2	N3	N5	N7	N8	N12	N13	N14	N18	N20	N25	N26	N28	N29	N30	N38	N40	U/S		
WCDMA	1	2	4	5	8															U/S	

Export Path: C:\WafToolData

Open Folder **Export**

Figure 4-55. Band Locked

2. Bands green indicates the currently configured bands of the module, and gray indicates the configurable bands.
3. When all of a system's band values are unchecked, you'll be prompted to click the **Set** button.
4. When the configuration is complete, click the **Set** button to complete the setup, and the RATs and bands content will be refreshed when the setup is successful;
5. The Factory Default for bands can be restored by clicking the **Restore** button, and the RATs and bands content is refreshed when set successfully.
6. Click **Open Folder** button to select a folder location. Then click **Export** button to export the bands information to an XML file.

5 Telephony Tool User Guide

The Telephony Tool serves as a unified interface for the previous Telephony tool on MTK platforms and the Debug Logging tool on Qualcomm platforms.

Currently, the Telephony Tool offers three main functionalities:

- Log capture
 - a. Collect System trace
 - b. Collect Host Driver trace (On windows OS only)
 - c. Collect modem trace
 - d. Collect dump log automatically
 - e. Monitor module crash (On windows OS only)
- Module FW Flash (Flash stress test function)
- User mode / Debug mode switch

5.1 The Main Interface

Each numbered marker identifies a specific item on the main page, and the explanations below detail the function or significance of each item as shown in the diagram.

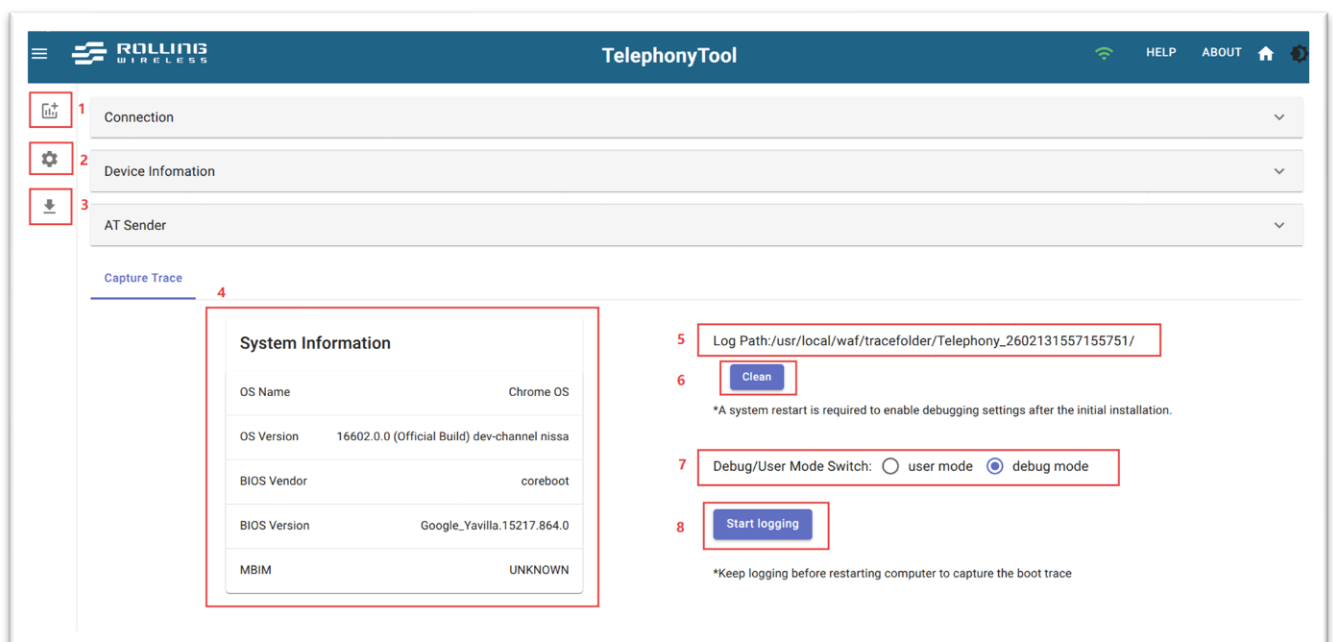


Figure 5-1. Main Interface of Telephony Tool

1. Icon to Capture Trace (Main Page)
Directs users to the main interface of the Telephony tool.
2. Icon to Trace Settings

Opens the trace settings configuration page.

3. Icon to Modem Flash

Navigates to the modem firmware flashing section.

4. System Information Display

Shows basic system details, including OS information and the status of the modem module.

5. Log File Destination

Displays the location where collected log files are saved.

6. Clear Collected Logs

A button to delete the collected log files; the "tracefolder" folder will be cleared.

7. User Mode / Debug Mode Switch

Allows users to toggle between user mode and debug mode.

8. Start / Stop Logging Button

Initiates or halts the logging process.

5.2 Capture Trace

5.2.1 User mode / Debug mode

The Telephony Tool provides a function that allows users to switch between User Mode and Debug Mode.

When switching between User Mode and Debug Mode, an operation result message will appear on the page to indicate the outcome.

Users can also check the updated port list in the Connection section—typically, more ports become available when the tool is in Debug Mode.



For the MTK platform, the NPT and ATT tools can only be run under debug Mode.

Please switch to Debug Mode and verify whether the MIPC port is available in the port list.

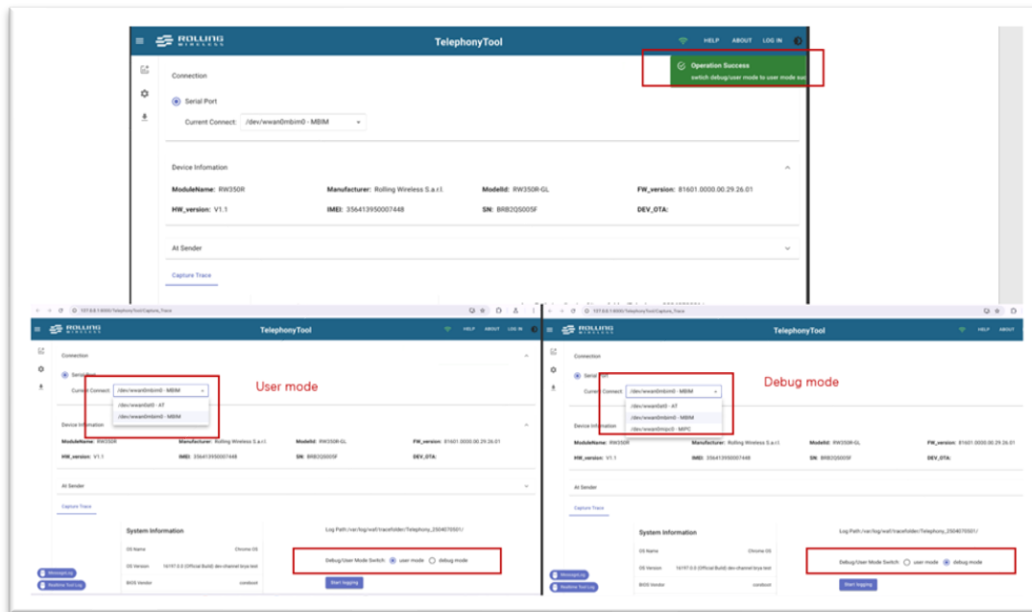


Figure 5-2. Switch between user mode and debug mode

However, the meaning and behavior of these modes may vary depending on the platform in use. The table below provides a detailed comparison to illustrate the differences across platforms.

Table 8. User mode / Debug mode / Platform mapping

Platform	Driver Interface	Mode type	Ports available	Modem Trace	NEMEA Data	Comment
Qualcomm Platform	USB	User mode	MBIM	No	No	63 mode
		Debug mode	MBIM AT Port Diag Port GNSS Port	Yes	Yes	42 mode
MTK Platform	PCIE	User mode	MBIM AT Port	Yes	No	--
		Debug mode	MBIM AT Port ADB Port MIPC Port GNSS Port	Yes	Yes	--
	USB	User mode	MBIM AT Port	No	No	--
		Debug	MBIM	Yes	Yes	--

mode

AT Port

ADB Port

MIPC Port

GNSS Port

5.2.2 Collect trace

5.2.2.1 Start / Stop logging

With the Telephony Tool, users can collect all relevant logs with a single click.

Once the **Start Logging** button is clicked, it will temporarily become disabled while the logging initialization process completes. After initialization, the button will switch to **Stop Logging**, accompanied by a spinning indicator to show that logging is actively in progress.

Users can click the **Stop Logging** button at any time to end the current logging session.

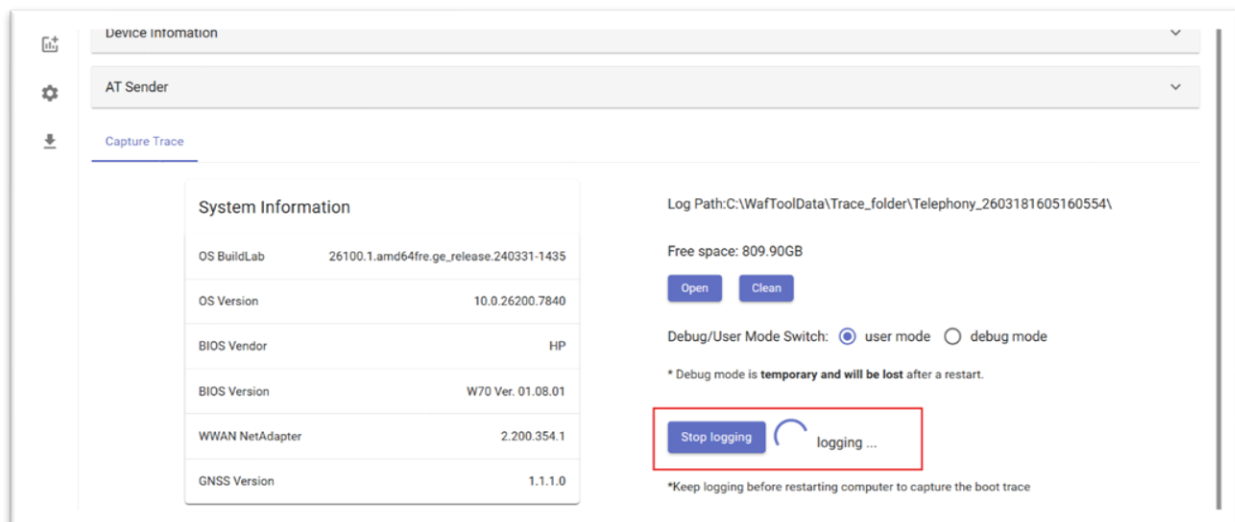


Figure 5-3. Start / Stop Logging



During the logging process, the status log file will continue to run, and it will not be deleted even if the WAF tool is uninstalled. The background process will also continue to collect logs. Therefore, If the tool is reinstalled during logging process, the UI interface will still display the logging status.

5.2.2.2 Log files

5.2.2.2.1 Log file storage location

All log files will be collected in the folder which shown in log path.

Users can access the log folder by clicking the **Open** button, and clear its contents using the **Clean** button.

The **Clean** button will permanently delete all contents within the "tracefolder" directory.



Please proceed with caution when using this option, as the action is irreversible.

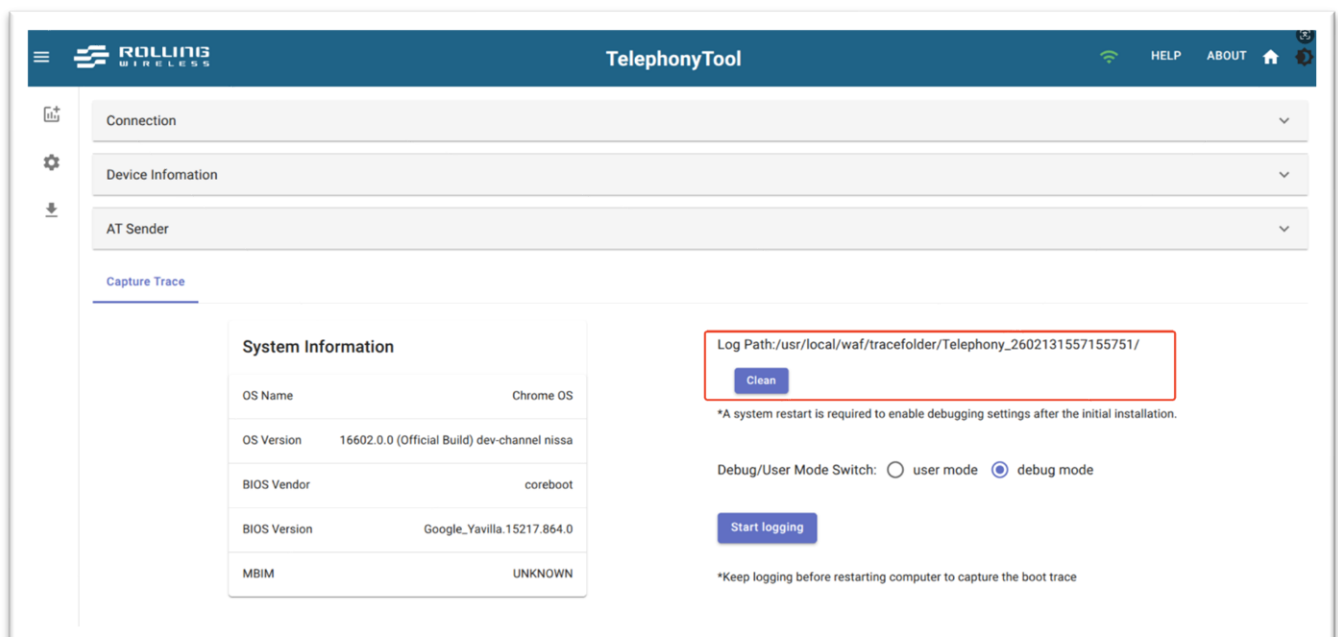


Figure 5-4. log file storage location



On Chrome OS, log files are copied to the user's "Downloads" folder, which is the default local storage directory accessible within the user's profile.

On Chrome OS, the Open button is not available. Users will need to manually navigate to the Downloads folder to access the saved log files.

Log files are organized into categorized folders for easier access and management:

- host – Contains all logs collected from the host side.
- MDlog – Stores modem trace logs.
- dump – Includes dump files automatically pulled through dump port or adb port.



1. On the Chrome OS MTK PCIe platform, initiating modem trace logging triggers an intentional modem reset by the tool. This behavior is by design and is required to capture complete modem trace information.
2. On Chrome OS, MTK PCIe platform dump files are pulled through the ADB port, which requires debug mode to be enabled. If the monitor auto-dump function is enabled, ensure that debug mode is re-enabled after each PC reboot.

Each type of log is saved in its corresponding folder to ensure clarity and facilitate post-processing or analysis.

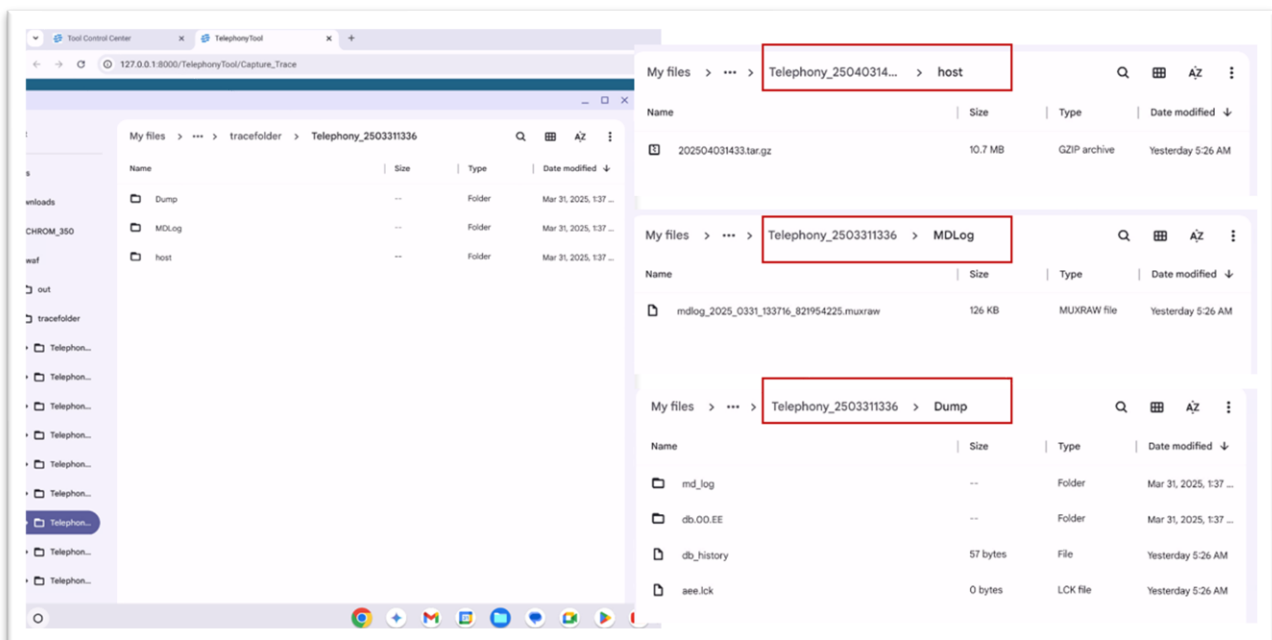
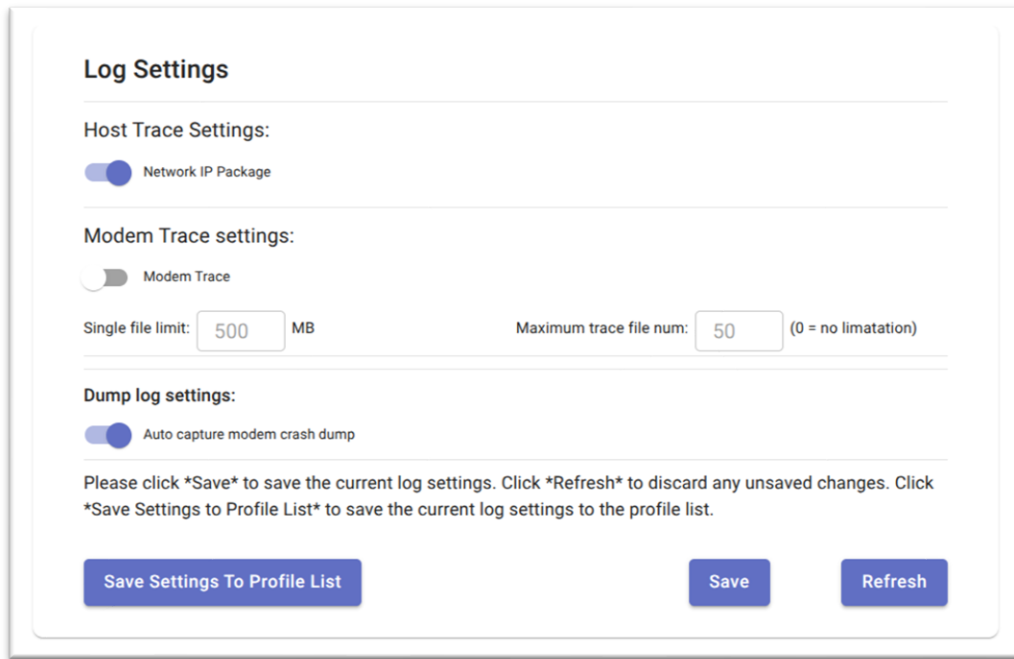


Figure 5-5. Chrome OS Log file directory structure

5.3 Trace settings

The Telephony Tool offers a user-friendly interface that allows users to easily customize how log files are handled.

These flexible options enable users to tailor log collection according to specific testing environments and workflows.



Log Settings

Host Trace Settings:

☒ Network IP Package

Modem Trace settings:

☐ Modem Trace

Single file limit: MB Maximum trace file num: (0 = no limitation)

Dump log settings:

☒ Auto capture modem crash dump

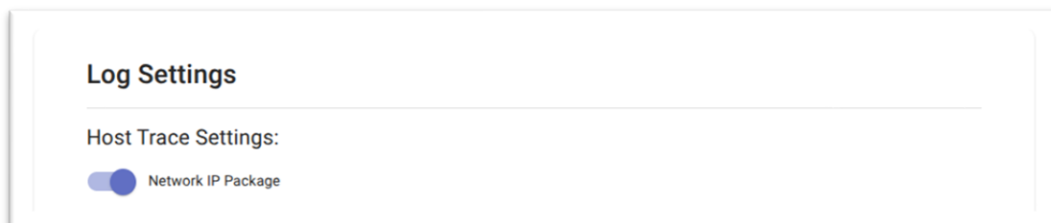
Please click *Save* to save the current log settings. Click *Refresh* to discard any unsaved changes. Click *Save Settings to Profile List* to save the current log settings to the profile list.

Figure 5-6. Main Interface of Trace Settings

5.3.1 Log file general setting

5.3.1.1 Host Trace Settings

The Telephony Tool also allows users to easily customize which types of logs are collected. Telephony tool can collect network Pcap log and system log.



Log Settings

Host Trace Settings:

☒ Network IP Package

Figure 5-7. Chrome OS host trace items

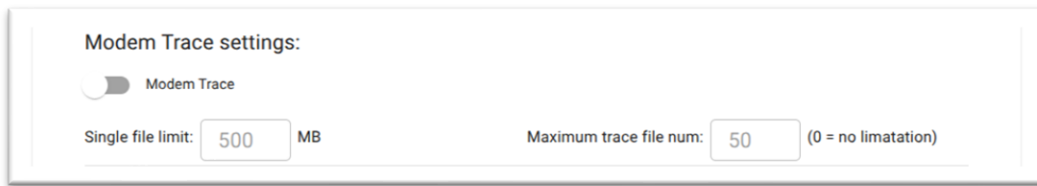
5.3.1.2 Modem Trace Settings

In general, we are not recommending to enable modem trace because modem logging brings heavy system loading, so Modem Trace is disabled by default. If needed, Modem Trace can be enabled to capture modem logs.

No internet access / network lag / No registration etc. issues, we need to capture Modem trace.

To capture modem trace log, make sure to check **Modem Trace ON** as below picture. After checked, you

can enter a file size limit and the number of trace files.



Modem Trace settings:

☐ Modem Trace

Single file limit: MB

Maximum trace file num: (0 = no limitation)

Figure 5-8. Modem trace settings

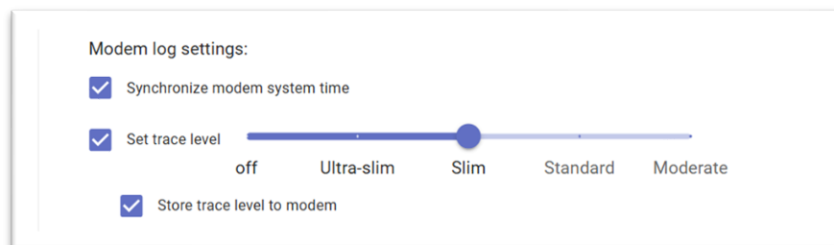


DO NOT enable modem trace during the following test cases:

- Stress testing
- CXP scenario related testing
- Power state related testing (S3/S4/MSR suspend/resume testing)
- Power performance related testing
- Driver installation/uninstallation related testing

5.3.1.2.1 Modem Log Settings

The Telephony Tool also provides a convenient UI for configuring modem trace-related settings.



Modem log settings:

☒ Synchronize modem system time

☒ Set trace level

off Ultra-slim Slim Standard Moderate

☒ Store trace level to modem

Figure 5-9. Modem Log Settings

Users can adjust the following options:

- **Synchronize Modem System Time**

Allows users to choose whether to synchronize the modem's internal clock with the system time. This is recommended for accurate log timestamp alignment.

- **Modem Trace Level Configuration (MTK platform only)**

Users can select from five trace level options to control the granularity of modem logging.

Note: The interpretation of each level may vary depending on the modem platform.

- Off
- Ultra Slim
- Slim

- Standard
- Moderate
- **Store Trace Level to Modem (MTK platform only)**

Users can choose whether to save the selected trace level to the modem itself, ensuring that the setting persists across sessions or reboots.

5.3.1.2.2 Dump Log Settings

The Telephony Tool can automatically capture modem crash dump logs, and provides UI options for configuring the modem dump type.

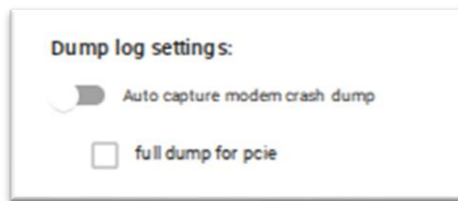


Figure 5-10. Dump log settings

- **Auto-Capture Modem Crash Dump**

Allows users to enable or disable automatic collection of modem dump logs when a crash occurs. For 135 module, if this option is selected, the module will automatically restart after capturing the log.

- **Modem dump type configuration (MTK platform only)**

Users can select the type of modem dump to be collected.

On Linux and Chrome OS, the full dump type is used for PCIe platforms to ensure the tool can retrieve dump logs via the ADB port.



If the modem dump type is changed via the UI, the new setting will not take effect until the modem is rebooted.

5.3.2 Save trace settings

After adjusting the settings on the Trace Settings page, users must click **Save** button to apply the changes.

Alternatively, the original settings can be restored at any time by clicking **Refresh** button.

An alert message is displayed upon saving the configuration to inform users of any important actions required, such as rebooting the modem for changes to take effect.

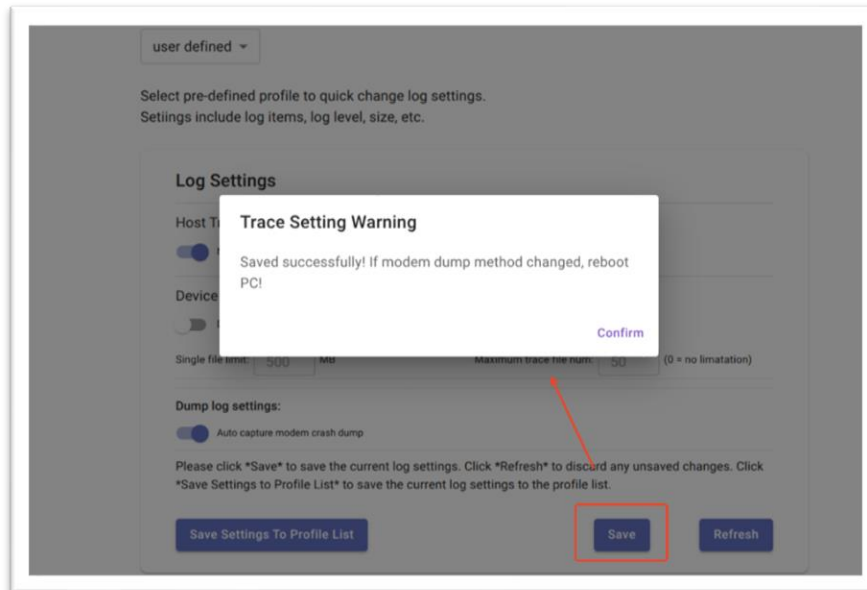


Figure 5-11. Trace settings save

Users can also save the current settings as a custom configuration template for future use.

This allows for quick re-application of preferred settings across different sessions or test scenarios.

By clicking **Save Settings to Profile List** and entering a file name, users can save the current configuration as a new profile.

The newly created profile will then appear in the profile list for easy selection and reuse.

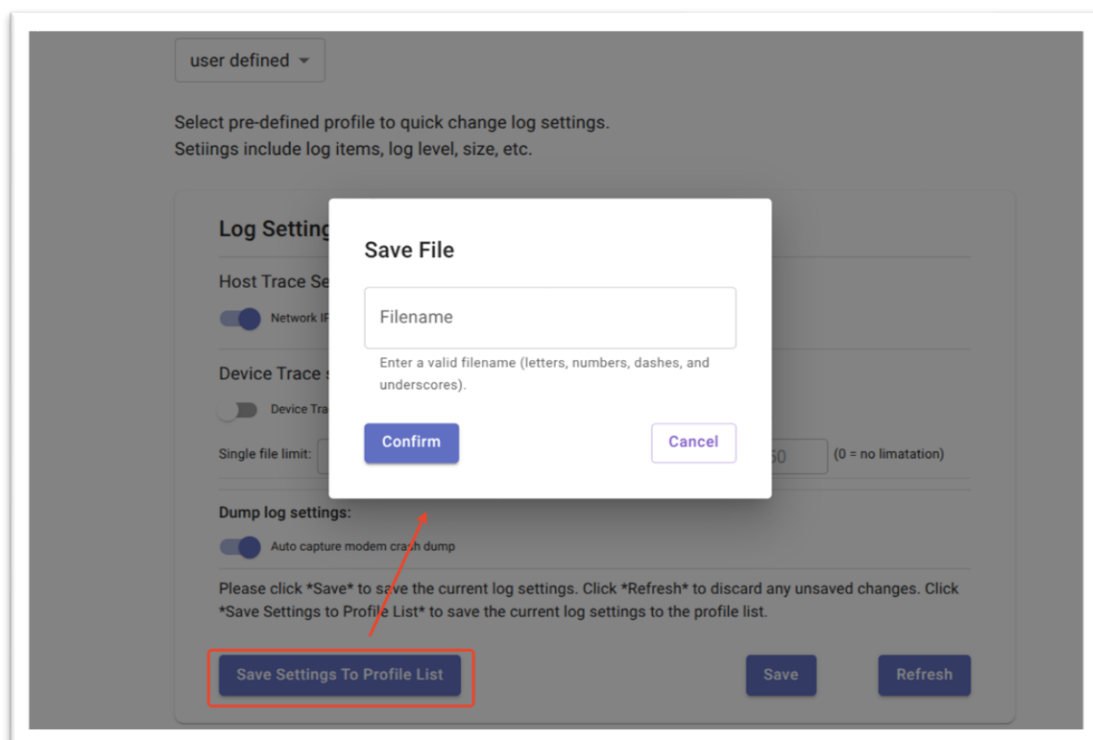


Figure 5-12. Save customized template

5.3.3 Fast start with loading Setting Template

The Telephony Tool provides several predefined configuration profiles to help users quickly adapt to different testing scenarios, as outlined below.

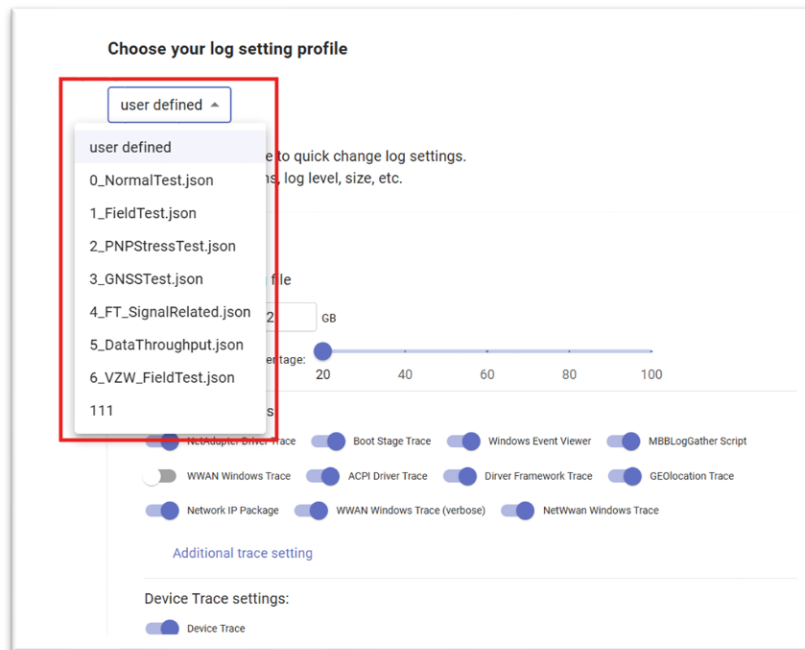


Figure 5-13. Pre-defined profile list

Currently, most of profiles are designed specifically for the MTK platform on Windows OS.

Users can select a suitable profile based on the test case requirements:

1. **NormalTest** – The default configuration used for DogFood testing. If there are no special requirements, this template is recommended. It shares the same logging items as the default settings.
2. **FieldTest** – Intended for field testing. This template enables additional trace logs. If modem trace is enabled, it does not modify the modem trace level.
3. **PNPStress** – Used for stress scenarios such as repeated S4 (suspend), resume, or reboot cycles.
4. **GNSSTest** – Tailored specifically for GNSS-related testing.
5. **FT_SignalRelated** – Designed for signal-related test cases during field testing. This template will automatically set the modem trace level to slim.
6. **DataThroughput** – Optimized for data throughput testing. It automatically sets the modem trace level using a predefined filter bin.



Users should follow the usage guidelines provided below when applying this profile.



On Linux and Chrome OS, any configuration templates that are not applicable to the current platform are automatically hidden from the user interface.

Table 9. Profile comparison

	0_NormalTest (DogFood):	1_FieldTest	2_PNPStressTest	3_GNSSTest	4_FT_SignalRe lated	5_DataThroughp ut
Netadapter Driver Trace	√	√	√	√	√	√
Boot Stage Trace	√	√	√	√	√	√
Windows Event Viewer	√	√	√	√	√	√
ACPI Driver Trace	√	√	√		√	√
MBBLogGat her Script	√	√	√	√	√	√
WWAN Windows Trace	√	√	√	√	√	√
Driver Framework Trace		√			√	√
Geolocation Trace		√		√		
Network IP Package						√
SAP log		√			√	√
Modem Trace		√			√	√
Monitor modem crash dump	√	√	√	√	√	√

	0_NormalTest (DogFood):	1_FieldTest	2_PNPStressTest	3_GNSSTest	4_FT_SignalRe lated	5_DataThroughp ut
Modem Trace level	Default (No modem trace level change)	Default (No modem trace level change)	Default (No modem trace level change)	Default (No modem trace level change)	Set trace level with MBIM command as Slim	Send a specially designed filter bin automatically at every MD trace startup
MSFT Log	Only activate them for special request					

5.4 Modem Flash

The Telephony Tool offers a modem firmware flashing function, along with support for flash stress testing.

After selecting the appropriate firmware file and erase mode, users can initiate the flashing process by clicking **Start** button.

The firmware will then be downloaded to the modem, and both the flashing progress and result will be displayed in the UI.

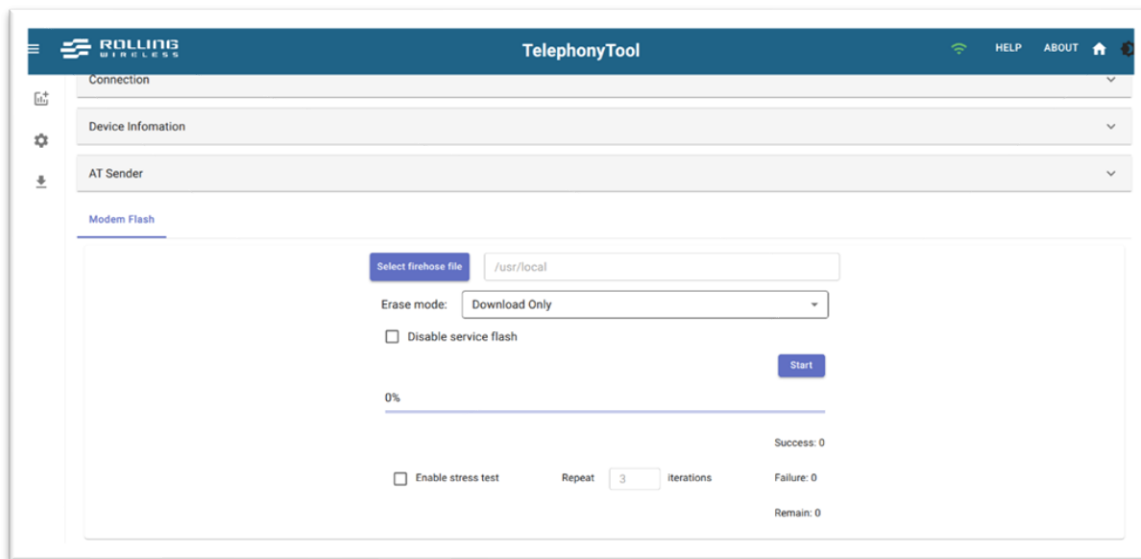


Figure 5-14. Modem Flash

The Telephony Tool also provides an additional function to support flashing operations:

- **Enable Stress Test**

By enabling stress test option, users can also perform flash stress testing to evaluate modem stability under repeated flashing scenarios.

Select firehose file

/usr/local

Erase mode: Download Only

☐ Disable service flash

Start

0%

☐ Enable stress test
 Repeat iterations
 Success: 0
 Failure: 0
 Remain: 0

Figure 5-15. Enable flash stress test

6 Noise Profiling Tool User Guide

6.1 Module Overview

The Noise Profiling Tool (NPT) is a tool used for scanning device internal noise. It utilizes the modem device's Rx measurement of Received Signal Strength Indicator (RSSI) to test the signal power received by each antenna of the modem under non-signaling mode. It can help us troubleshoot potential issues that may affect cellular mobile communication. Customers need to use this tool in a shielded environment in the laboratory, ensuring that the modem is installed internally, each antenna is properly connected and installed, and the device casing is intact. Due to limitations in modem functionality, the tool can only perform measurements within specific frequency bands and bandwidths, and the measurement results cannot fully replace spectrum analyzers.

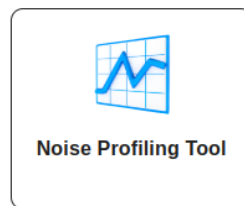


Figure 6-1. Function Button

6.2 Prerequisites

In 350 modules, NPT need to use MIPC port to do RF scan function. Please go to Telephony page to enable debug mode first.

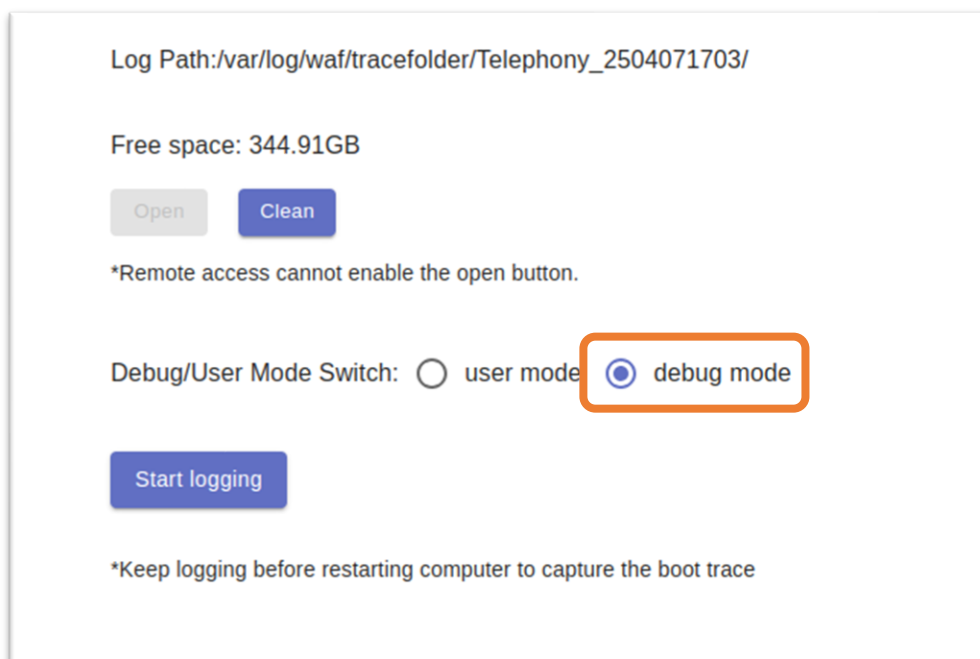


Figure 6-2. Switch MIPC port

After enable "debug mode", user should be able to find "wwan0mipc0" device under **Connection** section. User don't need to select MIPC port as "Current Connect", NPT will use this port in background.

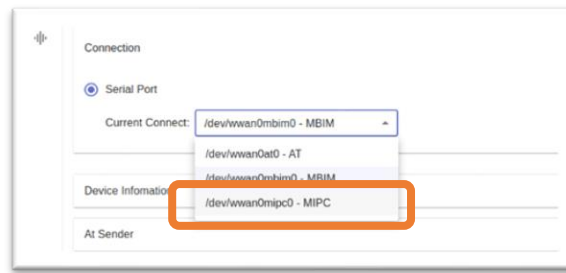


Figure 6-3. Connect MIPC port

6.3 Noise Scan Methods

NPT supports 3 different scan methods.

Table 10. Scan Method

Scan Method	Usage
Single Band Scan	Noise scan with one band configuration
Multi Bands Scan	Noise scan with multi bands configuration
Single Channel Scan	Continuously scan for noise on a frequency point.

6.3.1 Single Band Scan

To start a single band scan, user need to configure following RF parameter

Table 11. RF Parameter

RF Parameters	Role in Single Band Scan
RAT	Specifies the radio technology (e.g., LTE, NR)
Band	Specifies the frequency band
Rx/Tx BW	Bandwidth
DL Channel Start	Starting frequency of the downlink channel
DL Channel Stop	Stop frequency of the downlink channel
Scan Step	Scan step - indicates the increment in frequency between successive scan points
ANT Num	Antenna Number for scan
Expected Level	Represents the anticipated signal strength level.
Tx Ctrl	Controls Tx function during the scan
Tx RB Start	Marks the starting resource block for transmission
Tx RB Num	Specifies the number of resource blocks used for transmission
Tx MCS	Transmit Modulation and Coding Scheme
Tx Power	Sets the Tx power level

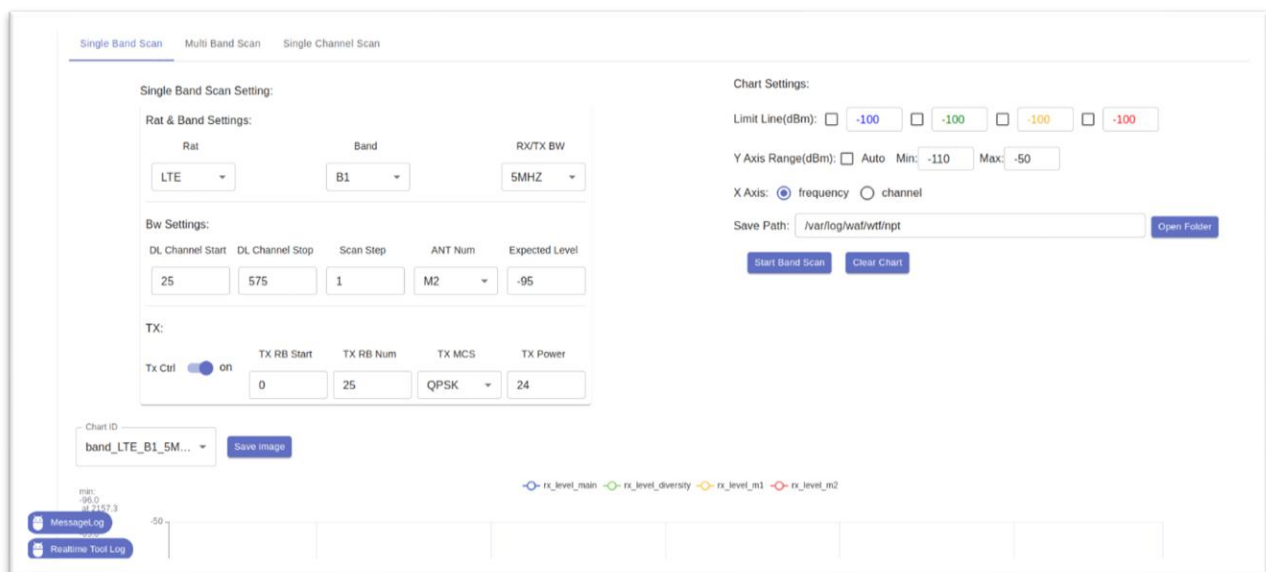


Figure 6-4. Single Band Scan

In the right side, there are some chart settings for noise chart drawing functions

Table 12. Chart settings for noise chart drawing

Setting Items	Function
Limit Line	Draw limit lines in the chart, support 4 customizable limit level with 4 colors
Y Axis setting	Y Axis is the noise level in dBm, customer to can set the Min and Max value for better display, or choose Auto setup based on measured noise level.
X Axis setting	X Axis can be set to frequency or channel number
Data Path	Data Path for report files and image files

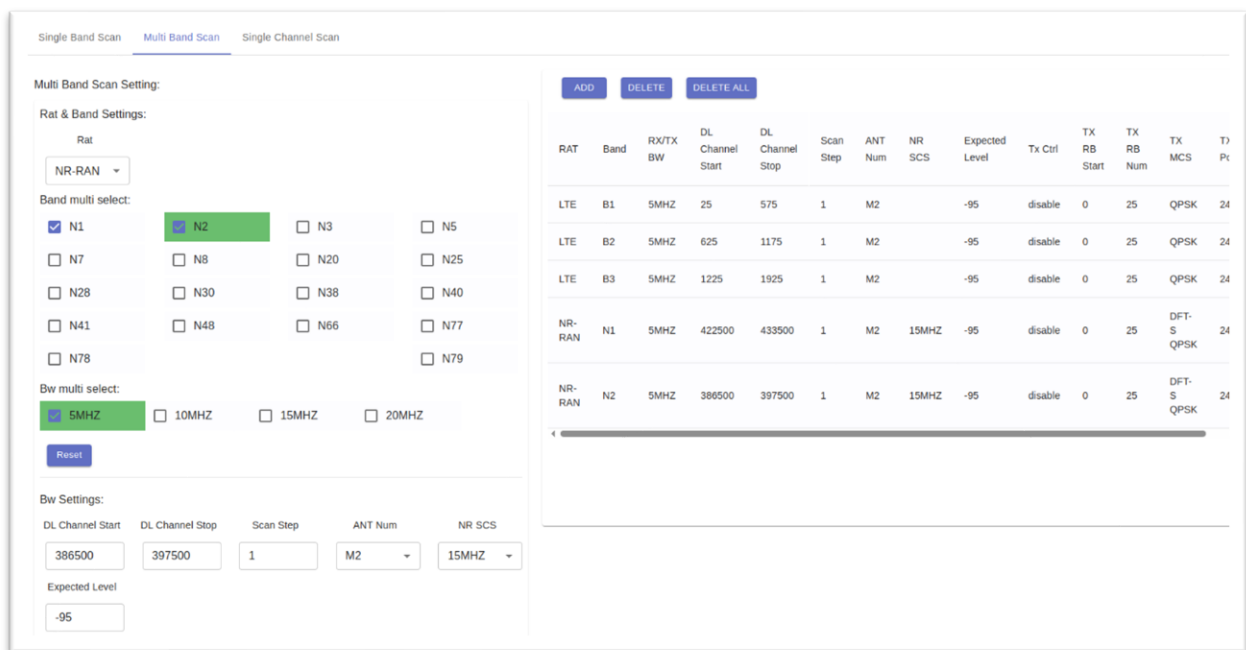
After selected RF parameters, please click "Start Band Scan" to start scanning.

The scan process will be stopped if,

- Scan channel reached stop channel
- User clicked Stop Scan
- Some Error happened during scan

6.3.2 Multi Bands Scan

User can use multi bands scan to configure a full frequency scan to cover all supported RATs and Bands.



RAT	Band	RX/TX BW	DL Channel Start	DL Channel Stop	Scan Step	ANT Num	NR SCS	Expected Level	Tx Ctrl	TX RB Start	TX RB Num	TX MCS	Tx Pr
LTE	B1	5MHZ	25	575	1	M2	-95	disable	0	25	QPSK	24	
LTE	B2	5MHZ	625	1175	1	M2	-95	disable	0	25	QPSK	24	
LTE	B3	5MHZ	1225	1925	1	M2	-95	disable	0	25	QPSK	24	
NR-RAN	N1	5MHZ	422500	433500	1	M2	15MHZ	-95	disable	0	25	DFT-S QPSK	24
NR-RAN	N2	5MHZ	386500	397500	1	M2	15MHZ	-95	disable	0	25	DFT-S QPSK	24

Figure 6-5. Multi Bands Scan

The RF parameters are same as single band scan, it allows you to add all the band and select multiple bandwidth and add it into a list which is displayed in right side.

1. Select Rat first

2. Select Bands need to be tested
3. In each Band, select the Bandwidths need to be tested
4. Click **Add** to Add items into test plan
5. Click **Start Multi Band Scan** to Start overall test



A complete full band - channel scan may take dozens of hours.

6.3.3 Single Channel Scan

Some time we may facing some random noise in specific frequency point. Single channel scan will do continuously scan for noise on a frequency point to help engineer to detect such kind of issue.

Compare to single band scan, there is no start/stop DL channel but only one DL channel.

There are two modes for Single Channel Scan. One is the timing mode, which measures the duration in minutes. The other is the counting mode, which measures the number of times.

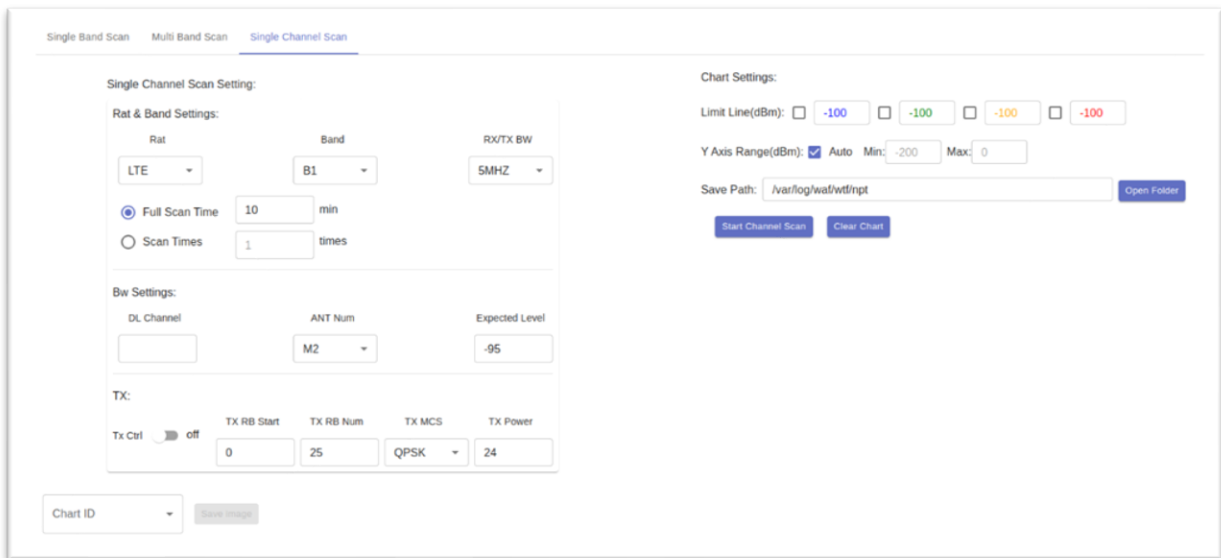


Figure 6-6. Single Channel Scan

6.4 Test Data and Report collection

After scan user can copy noise data file from saved data path.

If customer clicked **Save Image**, an image of the chart will be saved in the "image" folder under the same path as noise data files.



Figure 6-7. Report Collection

7 Antenna Tuner Tool User Guide

7.1 Interface Description

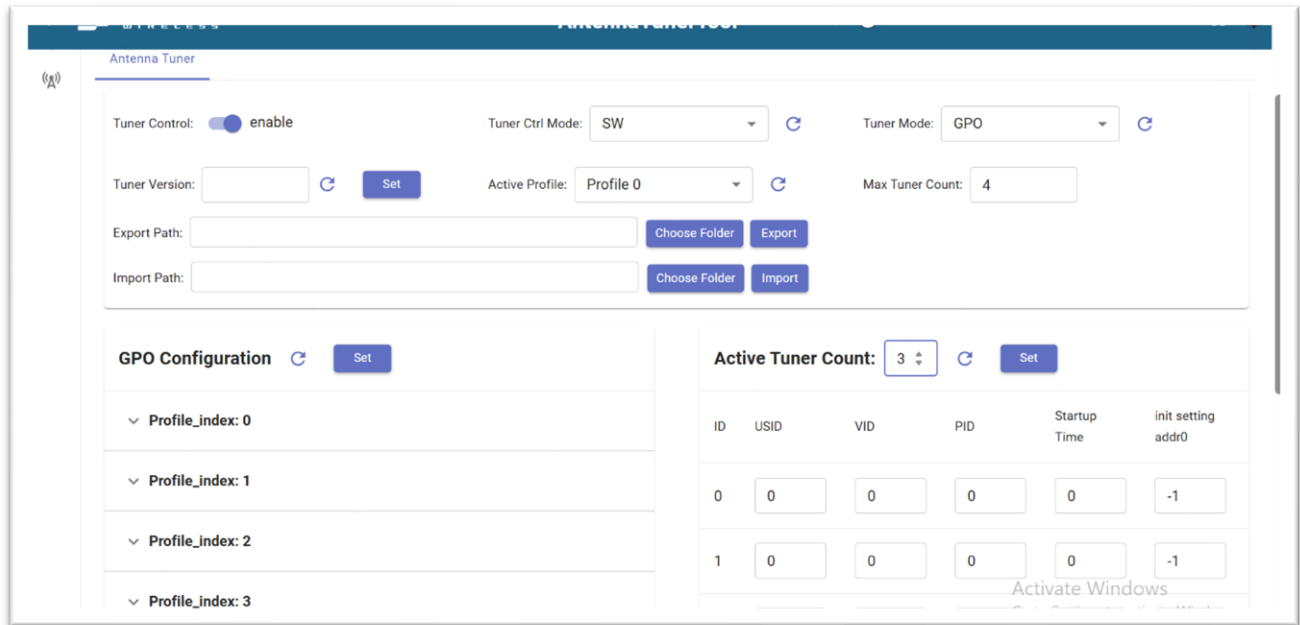


Figure 7-1. Interface Display

The AntennaTunerTool interface is composed of several functional sections, as shown in the figure:

1. Top Bar
 - a) Displays the tool title: AntennaTunerTool.
 - b) Shows the current tool version or branding (e.g., Rolling Wireless logo) if needed.
2. Modem Information Panel
 - a) Shows essential information such as Modem Model, FW Info.
 - b) Helps users confirm whether the device and firmware versions are recognized correctly.
3. Main Function Area
 - a) Provides controls for setting or viewing Tuner Control mode (e.g., Enable, Disable, HW mode, SW mode) and adjusting the Active Tuner Count.
 - b) Includes drop-down menus and input fields for selecting the Profile and performing actions such as Export, Import, and Apply.
4. MIPI Configuration Panel
 - a) Displays the current MIPI configuration parameters.
 - b) Allows users to edit fields (e.g., GD0, GD1) and apply changes to the module.
5. Active Tuner Count
 - a) Lists the status of each antenna tuning channel (Tuner) in either table or button form, making

it easy to view and manage multiple tuner configurations at once.

- b) Users can quickly see the current available number of tuners and perform enable/disable actions or parameter adjustments accordingly.

By organizing the user interface into these sections, the AntennaTunerTool makes it easy to configure MIPI parameters, switch tuner modes, and diagnose module communication in real time.

7.2 Enable and select the MIPC port

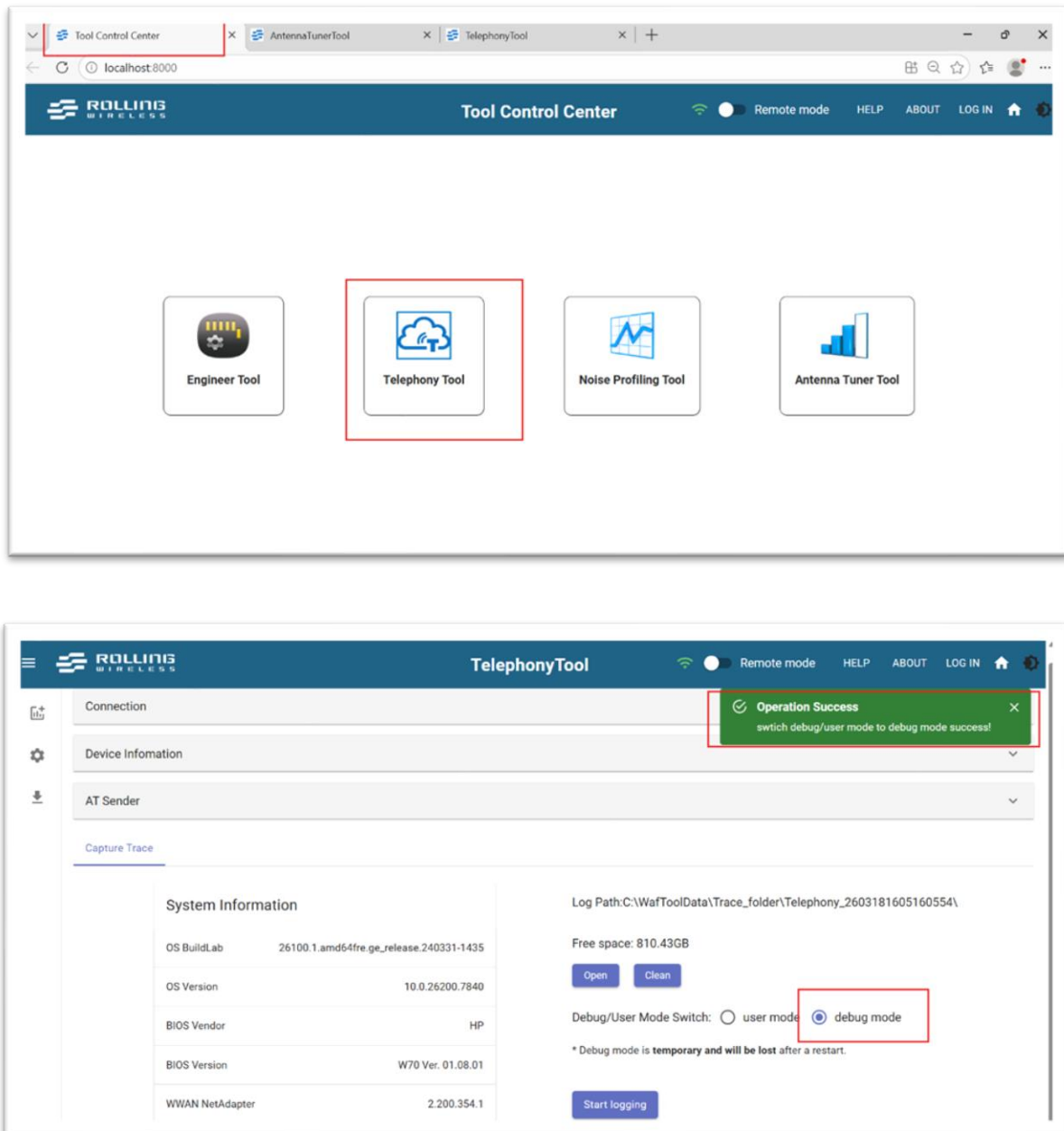


Figure 7-2. Enable the MIPC port

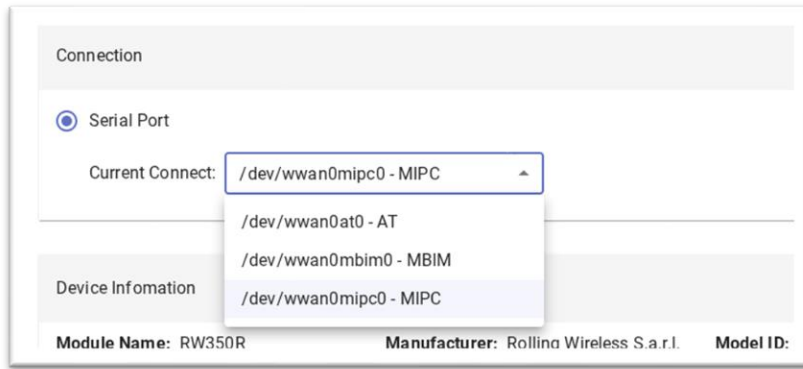


Figure 7-3. Select the MIPC port

1. Switch debug mode
 - a) On the main interface, click Telephony Tool.
 - b) Check the "debug mode" option.
2. Selecting the MIPC Port
 - a) After executing the above commands, return to the Antenna Tuner Tool interface.
 - b) Locate the newly enabled MIPI port in the connection options. The available port should now reflect the enabled status.
 - c) Select the MIPI port following the sequence shown in the reference image, and the tool will attempt to establish the connection.

Using these steps, you enable the MIPI port on your system and configure it for use with the tool. This process ensures that the MIPI interface is active and ready for transmitting the necessary data for antenna tuning configuration.

7.3 Tuner Configuration

After establishing a proper connection (e.g., selecting the MIPC port), proceed to the Tuner Configuration section to set up antenna tuning parameters. This chapter details each configuration option and the procedures for operating them.

7.3.1 Enable Tuner

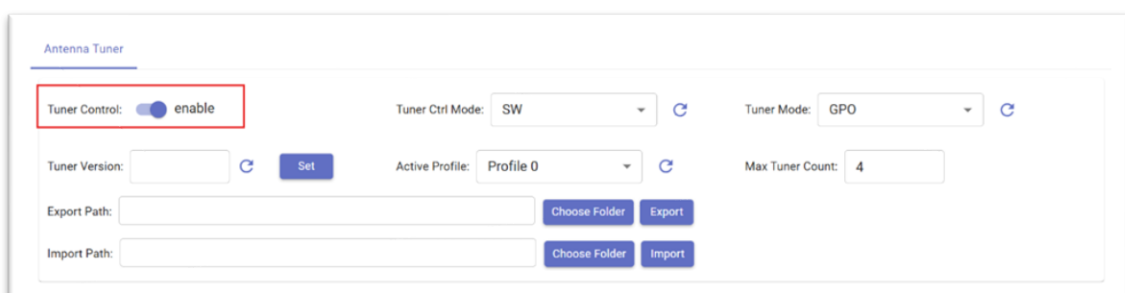


Figure 7-4. Enable Tuner

1. Locate the Tuner Control Toggle
 - a) Find the Tuner Control switch on the interface (as highlighted in the reference image).
 - b) By default, this option is set to Disable, which restricts access to certain tuner-related parameters.
2. Enable the Tuner
 - a) Click or toggle the Tuner Control button to change it to Enable.
 - b) Once enabled, tuner-related options—such as Tuner Mode (HW/SW), Tuner Count, and Active Profile settings—will become available for configuration.



If enabling Tuner Control fails, verify that the drivers, device connections, or pre-debug configurations (such as setting `t7xx_debug_ports`) are correctly applied.
The default value is Disable.

7.3.2 Switch to SW Control Mode

After the user enables the tuner operation, the user can choose the tuner control mode. As shown below.

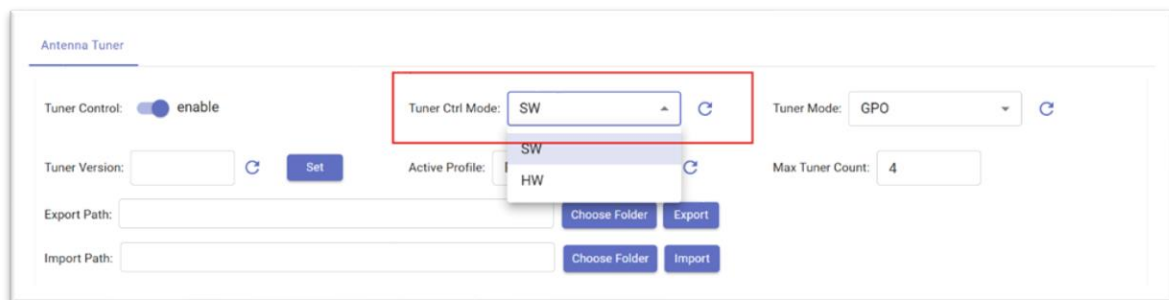
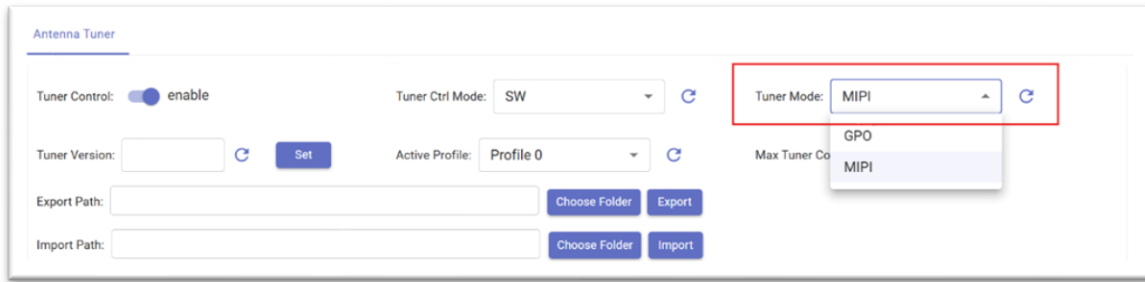


Figure 7-5. Switch to SW Control

Currently, only two tuner control mode are support, they are: "HW mode" and "SW mode".

7.3.3 Choose the MIPI Mode

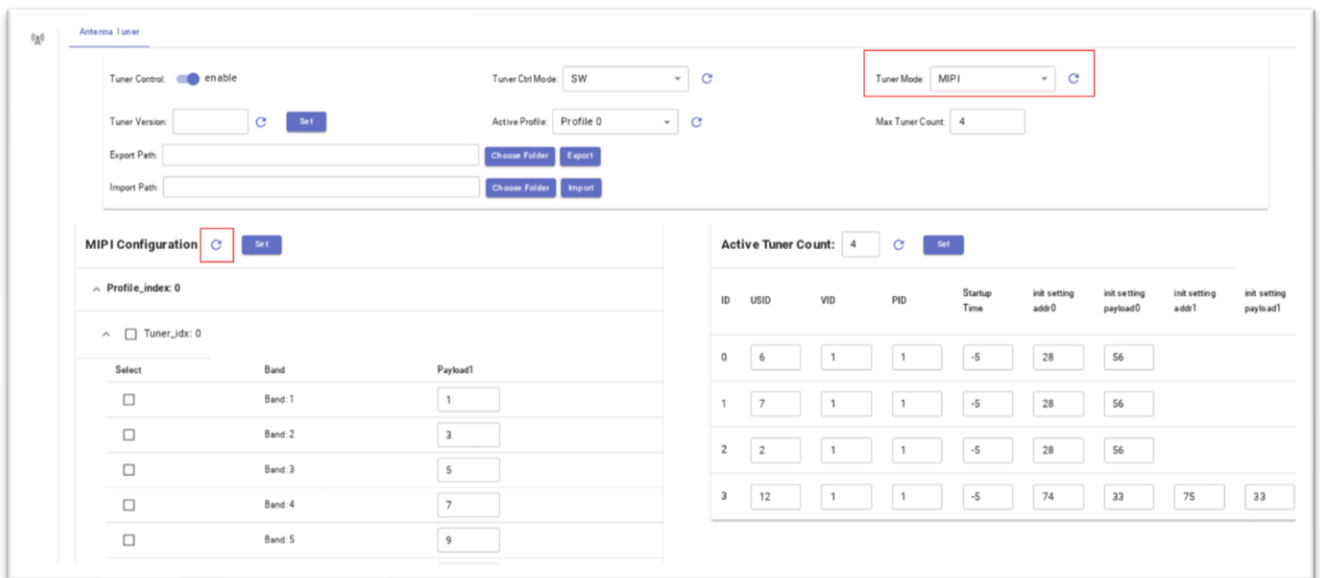
After the user choose the tuner control mode operation, the user can choose the tuner mode, as shown below.



The screenshot shows the 'Antenna Tuner' configuration window. The 'Tuner Mode' dropdown menu is open, showing 'MIPI' as the selected option. Other options visible are 'GPO' and 'MIPI'. The 'Tuner Control' is enabled, 'Tuner Ctrl Mode' is 'SW', 'Active Profile' is 'Profile 0', and 'Max Tuner Count' is 4.

Figure 7-6. Choose the MIPI

After the user choose the tuner MIPI mode operation, user can get one choose profile MIPI configuration from module. As shown below.



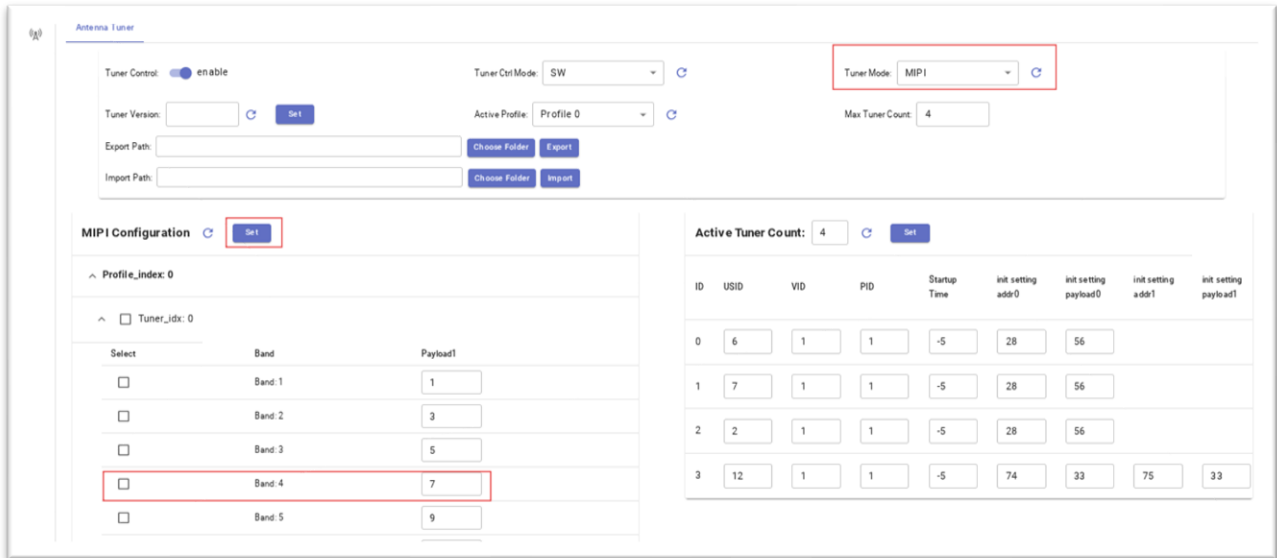
The screenshot shows the 'Antenna Tuner' configuration window with the 'MIPI Configuration' section expanded. The 'Tuner Mode' is set to 'MIPI'. The 'Active Tuner Count' is 4. The 'MIPI Configuration' section shows a table for selecting profiles based on Band and Payload1.

ID	USID	VID	PID	Startup Time	init setting add0	init setting payload0	init setting add1	init setting payload1
0	6	1	1	-5	28	56		
1	7	1	1	-5	28	56		
2	2	1	1	-5	28	56		
3	12	1	1	-5	74	33	75	33

Figure 7-7. Select profile MIPI Configuration

7.3.4 Set MIPI Configuration

After the user get one profile MIPI configuration, user can change some configuration value in this tool, and click the **Set** button to make it validate. As shown below.



Antenna Tuner

Tuner Control: ☒ enable

Tuner Ctrl Mode: SW

Tuner Mode: MIPI

Tuner Version: Set

Active Profile: Profile 0

Max Tuner Count: 4

Export Path: Choose Folder Export

Import Path: Choose Folder Import

MIPI Configuration Set

Profile_index: 0

Tuner_idx: 0

Select	Band	Payload1
☐	Band: 1	1
☐	Band: 2	3
☐	Band: 3	5
☒	Band: 4	7
☐	Band: 5	9

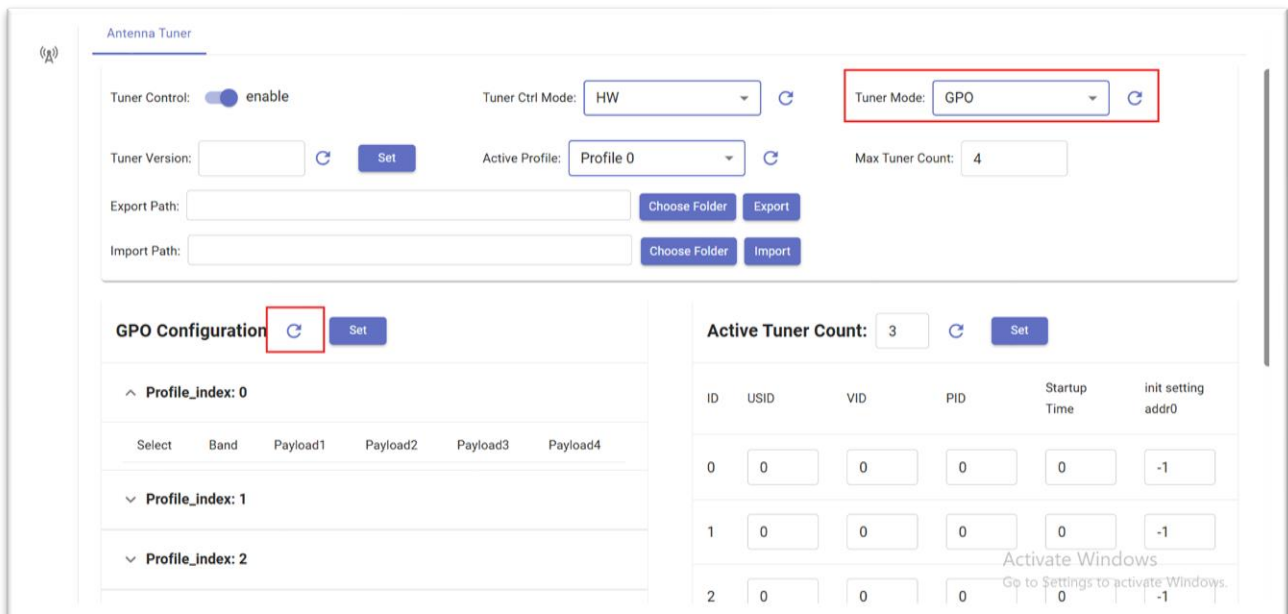
Active Tuner Count: 4 Set

ID	USID	VID	PID	Startup Time	init setting addr0	init setting payload0	init setting addr1	init setting payload1
0	6	1	1	-5	28	56		
1	7	1	1	-5	28	56		
2	2	1	1	-5	28	56		
3	12	1	1	-5	74	33	75	33

Figure 7-8. Set MIPI Configuration

7.3.5 Get GPO Configuration

After the user choose the tuner GPO mode operation, user can get one choose profile MIPI configuration from module. As shown below.



Antenna Tuner

Tuner Control: ☒ enable

Tuner Ctrl Mode: HW

Tuner Mode: GPO

Tuner Version: Set

Active Profile: Profile 0

Max Tuner Count: 4

Export Path: Choose Folder Export

Import Path: Choose Folder Import

GPO Configuration Set

Profile_index: 0

Profile_index: 1

Profile_index: 2

Select	Band	Payload1	Payload2	Payload3	Payload4
☐	Band: 1	1			
☐	Band: 2	3			
☐	Band: 3	5			
☒	Band: 4	7			
☐	Band: 5	9			

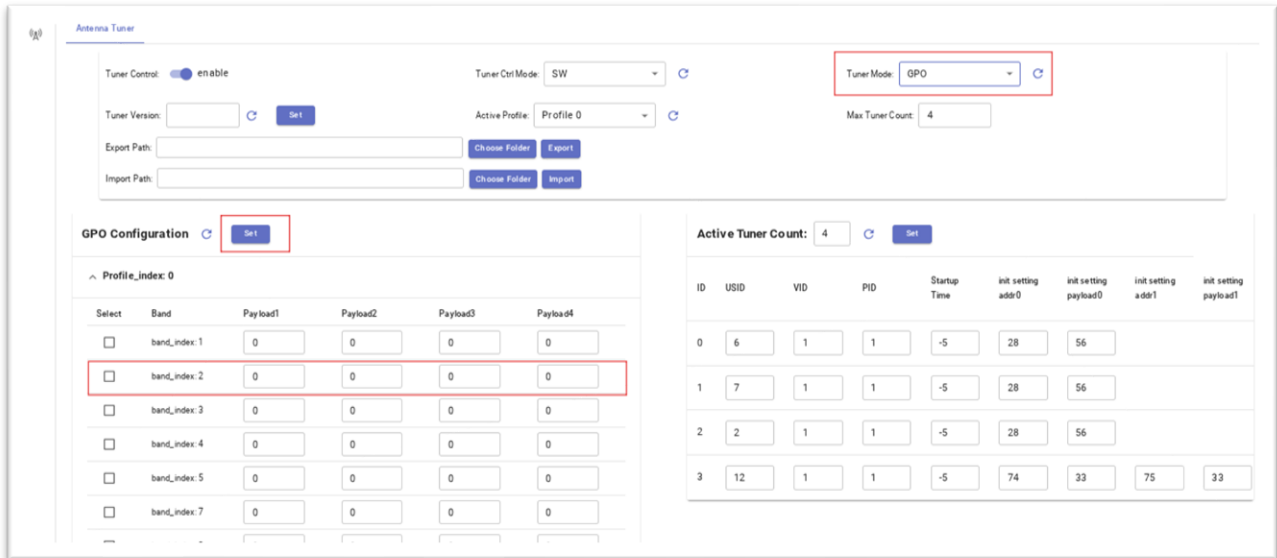
Active Tuner Count: 3 Set

ID	USID	VID	PID	Startup Time	init setting addr0
0	0	0	0	0	-1
1	0	0	0	0	-1
2	0	0	0	0	-1

Figure 7-9. Get GPO Configuration

7.3.6 Set GPO Configuration

After the user get one profile GPO configuration, user can change some configuration value in this tool, and click the **Set** button to make it validate. As shown below.



Antenna Tuner

Tuner Control: ☒ enable

Tuner Ctrl Mode: SW

Tuner Mode: GPO

Tuner Version: Set

Active Profile: Profile 0

Max Tuner Count: 4

Export Path: Choose Folder Export

Import Path: Choose Folder Import

GPO Configuration Set

Profile_index: 0

Select	Band	Payload1	Payload2	Payload3	Payload4
☐	band_index: 1	0	0	0	0
☒	band_index: 2	0	0	0	0
☐	band_index: 3	0	0	0	0
☐	band_index: 4	0	0	0	0
☐	band_index: 5	0	0	0	0
☐	band_index: 7	0	0	0	0

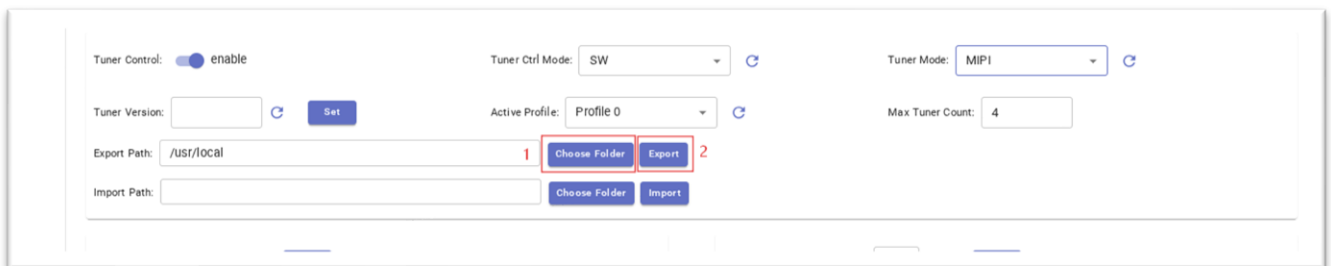
Active Tuner Count: 4 Set

ID	USID	VID	PID	Startup Time	init setting addr0	init setting payload0	init setting addr1	init setting payload1
0	6	1	1	-5	28	56		
1	7	1	1	-5	28	56		
2	2	1	1	-5	28	56		
3	12	1	1	-5	74	33	75	33

Figure 7-10. Set GPO Configuration

7.3.7 Exporting Current Profile MIPI Configuration

User can click the **Export** button to export current profile MIPI configuration. As show below.



Antenna Tuner

Tuner Control: ☒ enable

Tuner Ctrl Mode: SW

Tuner Mode: MIPI

Tuner Version: Set

Active Profile: Profile 0

Max Tuner Count: 4

Export Path: /usr/local 1 Choose Folder Export 2

Import Path: Choose Folder Import

Figure 7-11. Export current Profile MIPI Configuration



Due to limitations in the module chip platform, the currently supported export table range is: 0~8. The numerical value represents the subscript

The format of the XML file is shown in the following figure.

```

root@uldrenite-rev1:/var/log/waf/ # cat AntFile_Table_2.xml
</Antenna_Tuner>uldrenite-rev1 /var/log/waf/wtf/att # cat AntFile_Table_2.xml
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<!--This is Antenna Tuner Tool Export xml file Format.-->
<Antenna_Tuner>
  <Antenna_Table_Ver></Antenna_Table_Ver>
  <Antenna_Tuner_Type>
    <Type name="GP0" TunerRegisterNum="4"/>
  </Antenna_Tuner_Type>
  <Antenna_Tuner_Cfg>
    <Tuner TunerId="0" TunerUsid="6" PmTrigger="0x01" PayloadRegisterAddress="0x01" Star
    <Tuner TunerId="1" TunerUsid="7" PmTrigger="0x01" PayloadRegisterAddress="0x01" Star
    <Tuner TunerId="2" TunerUsid="2" PmTrigger="0x01" PayloadRegisterAddress="0x01" Star
    <Tuner TunerId="3" TunerUsid="12" PmTrigger="0x01" PayloadRegisterAddress="0x01" Sta
  </Antenna_Tuner_Cfg>
  <Antenna_Tuner_Data>
    <Data TunerId="0">
      <Band name="Band17" GD0="224" GD1="0" GE0="0" GE1="0"/>
      <Band name="Band18" GD0="194" GD1="0" GE0="0" GE1="0"/>
      <Band name="Band26" GD0="194" GD1="0" GE0="0" GE1="0"/>
      <Band name="Band29" GD0="224" GD1="0" GE0="0" GE1="0"/>
      <Band name="Band32" GD0="224" GD1="0" GE0="0" GE1="0"/>
      <Band name="Band34" GD0="0" GD1="0" GE0="0" GE1="0"/>
      <Band name="Band42" GD0="0" GD1="0" GE0="0" GE1="0"/>
      <Band name="Band43" GD0="0" GD1="0" GE0="0" GE1="0"/>
      <Band name="Band46" GD0="0" GD1="0" GE0="0" GE1="0"/>
      <Band name="Band77" GD0="0" GD1="0" GE0="0" GE1="0"/>
      <Band name="Band78" GD0="0" GD1="0" GE0="0" GE1="0"/>
      <Band name="Band79" GD0="0" GD1="0" GE0="0" GE1="0"/>
    </Data>
    <Data TunerId="1">
      <Band name="Band17" GD0="8" GD1="0" GE0="0" GE1="0"/>
      <Band name="Band18" GD0="0" GD1="0" GE0="0" GE1="0"/>
      <Band name="Band26" GD0="0" GD1="0" GE0="0" GE1="0"/>
      <Band name="Band29" GD0="8" GD1="0" GE0="0" GE1="0"/>
      <Band name="Band32" GD0="0" GD1="0" GE0="0" GE1="0"/>
      <Band name="Band34" GD0="0" GD1="0" GE0="0" GE1="0"/>
    </Data>
  </Antenna_Tuner_Data>
</Antenna_Tuner>

```

Figure 7-12. Exported MIPI Configuration

7.3.8 Importing Current Profile MIPI Configuration

User can click the **Import** button to export current profile MIPI configuration. As shown below.

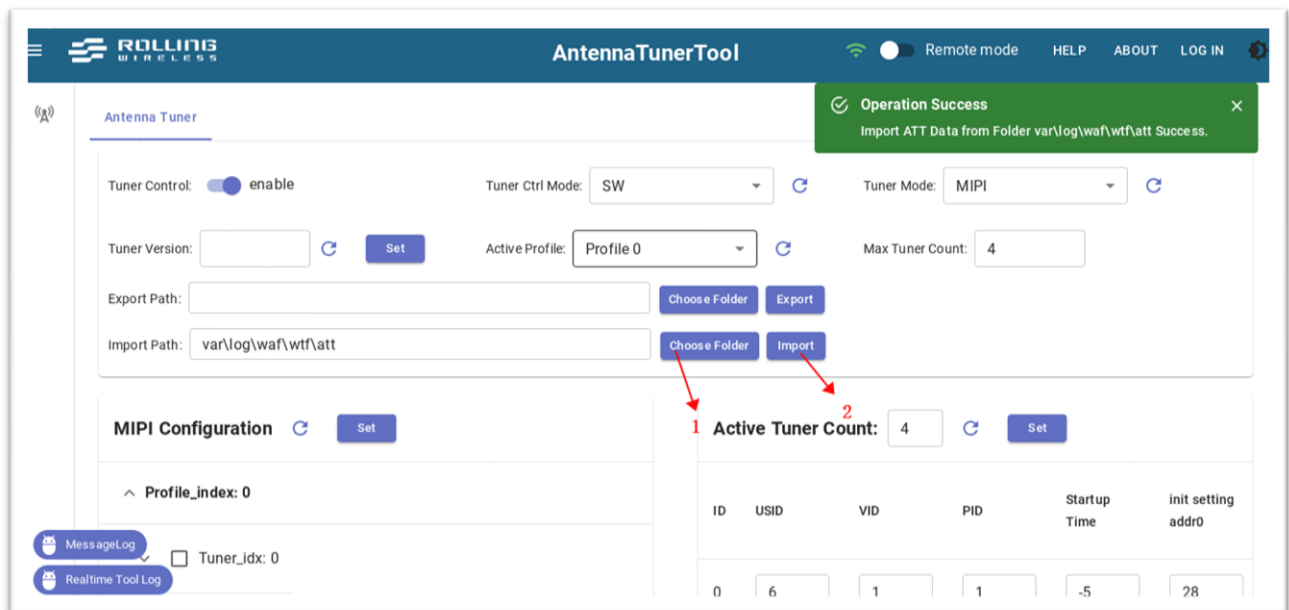
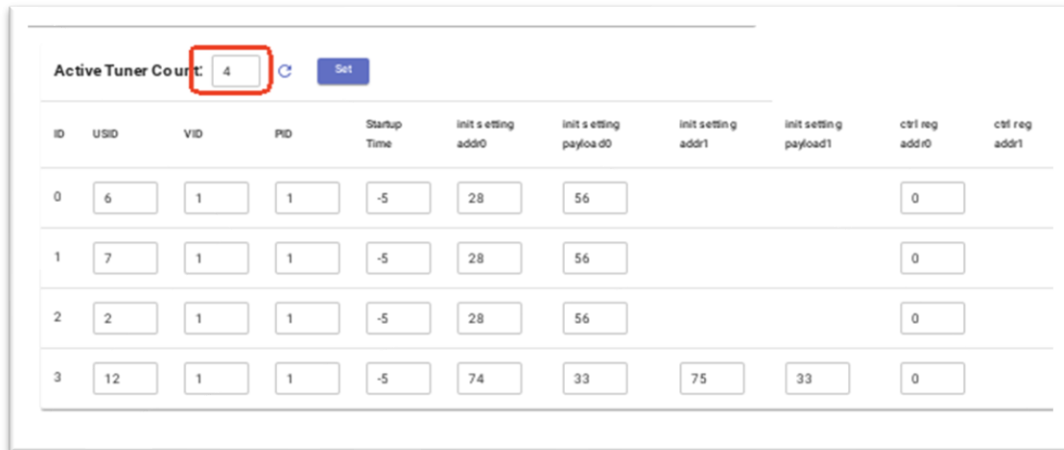


Figure 7-13. Import MIPI Configuration

7.3.9 Get Tuner IC Configuration

After the user choose the tuner count, user can get Tuner IC configuration from module. As shown below.

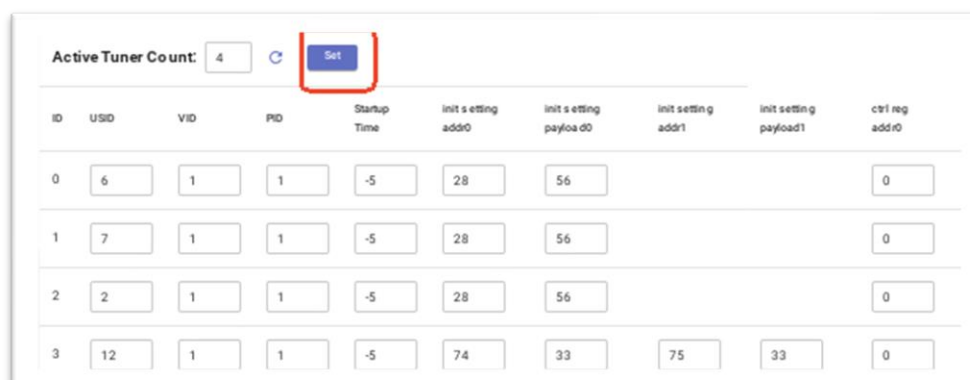


ID	USID	VID	PID	Startup Time	init setting addr0	init setting payload0	init setting addr1	init setting payload1	ctrl reg addr0	ctrl reg addr1
0	6	1	1	-5	28	56			0	
1	7	1	1	-5	28	56			0	
2	2	1	1	-5	28	56			0	
3	12	1	1	-5	74	33	75	33	0	

Figure 7-14. Get Tuner IC configuration

7.3.10 Set Tuner IC Configuration

After the user get Tuner IC configuration, user can change some configuration value in this tool, and click the **Set** button to make it validate. As shown below.



ID	USID	VID	PID	Startup Time	init setting addr0	init setting payload0	init setting addr1	init setting payload1	ctrl reg addr0	ctrl reg addr1
0	6	1	1	-5	28	56			0	
1	7	1	1	-5	28	56			0	
2	2	1	1	-5	28	56			0	
3	12	1	1	-5	74	33	75	33	0	

Figure 7-15. Set Tuner IC configuration

8 Troubleshooting & Support

8.1 FAQ (Frequently Asked Questions)

N/A

8.2 Useful scenarios

N/A

8.3 Other suggestions

N/A